"Relative Histories" Formulation of Quantum Mechanics I: Introduction

W. David Strayhorn, MD, PhD*
Department of Neurology, Washington University School of Medicine, Saint Louis, MO 63110
(Dated: January 5, 2005)

A novel formulation of quantum mechanics, the "relative histories" formulation (RHF), is proposed based upon the assumption that the physical state of a closed system – e.g., the universe – can be represented mathematically by an ensemble E of four-dimensional manifolds $\mathcal{W}$. In this scheme, any real, physical object – be it macroscopic or microscopic – can be assigned to the role of the quantum mechanical "observer." The state of the observer is represented as a 3-dimensional manifold, $\mathcal{O}$. By using $\mathcal{O}$ as a boundary condition, we obtain E as the unique set of all $\mathcal{W}$ that satisfy the boundary condition defined by $\mathcal{O}$. The evolution of the "wavefunction of the universe" E is determined by the movement of the observer $\mathcal{O}$ through state space. The development of this scheme is guided by the desire to establish a formulation with the following interpretational features: (1) a clear definition of the observer and of its space of states, as well as the lack of the fundamental split between observer and system that is characteristic of the Copenhagen Interpretation; (2) movement of the observer through state space that obeys classical notions of locality and probability; (3) a derivation of quantum statistics and quantum nonlocality from classical notions of probability and locality; (4) a demonstration of compatibility with general relativity; (5) an adherence to the notion that "all is geometry;" (6) the absence of a requirement for any extra "unphysical" dimensions of spacetime beyond the four of our everyday experience; and (7) a spacetime structure that is causal on the large scale. In addition, the RHF provides a simple conceptualization of the EPR argument that QM cannot be considered both local and complete. In exchange for these interpretational features, we are forced merely to accept that the structure of spacetime is acausal *on the small scale*. To demonstrate the feasibility of the RHF as a formulation of quantum mechanics, we argue that the RHF and the Feynman path integral (FPI) approach always make the same predictions. This correspondence between the RHF and the FPI exists despite the fact that quantum statistics per se is not explicitly assumed by the RHF. The RHF, therefore, is a fundamentally classical theory that is grounded in the same mathamatical language as general relativity and, as such, is proposed as a theory of quantum gravity.

PACS numbers: PACS numbers go here. Abbreviations: RHF, relative histories formulation; QM, quantum mechanics; GR, general relativity; FPI, Feynman path integral; CPC, chronology protection conjecture.

I. INTRODUCTION

This paper presents an outline of a novel formulation of quantum mechanics referred to as the "relative histories" formulation (RHF) of quantum mechanics. The justification for the development of the RHF is twofold. First, the RHF offers a constellation of *interpretational* advantages that is not found in any other formulation of quantum mechanics. Second, the RHF provides a vision of a common mathematical framework for the development of general relativity as well as quantum mechanics. As such, the RHF is an attempt at a theory of quantum gravity.

The purpose of the present work is to present the basics of the RHF and to demonstrate explicitly that it makes the same predictions as any other formulation of quantum mechanics. In a companion paper [19], the interpretational advantages of the RHF will be discussed in some detail. In a second companion paper [20] (in progress), the mathematical underpinnings of the RHF will be explored and developed in greater detail.

The outline of the present paper is as follows. In Sec. II, the basic framework of the RHF is presented through an introduction to our definition and use of manifolds, the mathematical representation of the observer, the state space of the observer, the movement of the observer through state space, and the rules of probability that govern this movement. In Sec. III, we present a model of elementary particles as topologically nontrivial structures in the fabric of four dimensional spacetime, the so-called "4-geon" particle model introduced by Hadley [8]. We will make use of the framework introduced in Sec. II and Sec. III in Sec. IV, in which we will demonstrate that the formalism of the FPI and the formalism of the Feynman path integral (FPI) technique are equivalent. In other words, we will argue that the RHF and the FPI make the same predictions. Since the FPI is well-known [1] to provide a basis for derivation of the Schrödinger equation, this equivalence between the RHF and the FPI implies that the RHF is equivalent to quantum mechanics. The RHF therefore qualifies as an independent formulation of quantum mechanics. In the Discussion, we summarize the postulates of the RHF and discuss its further development.

II. BASIC FRAMEWORK

Let us begin with an overview of the basic framework of the RHF. The "center stage" of the RHF is occupied by the observer. In principle, we make the assumption that *any real, physical object* can play the role of the observer. This can be macroscopic, like a data recording device, or microscopic, like a particle. The physical state of the observer at any given

*Electronic address: Wstrayho@im.wustl.edu, straycat_md@yahoo.com

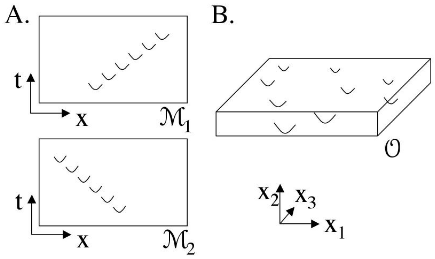

FIG. 1: A. Two 2-dimensional manifolds $\mathcal{M}_1$ and $\mathcal{M}_2$ that are not diffeomorphic to one another must represent different overall physical realities. The U-shaped curves in the figure represent small scale deformations in the fabric of spacetime induced by a moving particle. In $\mathcal{M}_1$, the particle is moving in the positive-x direction; in $\mathcal{M}_2$, it is moving in the negative-x direction. B. My computer can be represented mathematically as the three dimensional object manifold $\mathcal{O}$. Each U-shaped curve corresponds to a particle. (Only nine of the gazillion or so particles that make up my computer are depicted.)

point in its evolution will be represented by a single *three* dimensional manifold, $\mathcal{O}$. (We will see in [20] that $\mathcal{O}$ is an element of an infinite-dimensional state space.) In the framework of the RHF, the physical state of any object other than the observer – including the physical state of the Universe itself – is defined explicitly in terms of its relationship to the observer. That is, the state of the Universe is represented as an ensemble E (a set) of individual *four* dimensional manifolds $\mathcal{W}$. E is defined explicitly as a function of $\mathcal{O}$, that is, as the unique set of all $\mathcal{W}$ that satisfy the boundary condition defined by $\mathcal{O}$. The time-dependent evolution of E is therefore determined by the movement of $\mathcal{O}$ through state space. (Note an important distinction: the observer is defined as a *single* manifold $\mathcal{O}$, which in turn defines an entire *ensemble* of manifolds, $\mathcal{W}$.) In a sense, the movement of $\mathcal{O}$ through state space can be conceptualized by analogy to the Brownian motion movement of a dust particle through the air. The relevant similarities are that both types of movement through their respective state spaces follow classical, intuitive notions of probability and locality. This will hopefully become more clear below with the introduction of the "random walk hypothesis" governing the motion of the observer through state space. In the following subsections, we will develop these concepts more fully.

A. Manifolds

As just stated, our starting assumption is the notion that the state of the Universe can be described in terms of an ensemble E (a set) of individual four-dimensional manifolds $\mathcal{W}$, each with a prescribed topology W as well as a metric g_{ab} defined over the manifold. (Thus, $\mathcal{W}$ will be used to indicate a particular topology as well as a particular metric: (W, g_{ab}).) These manifolds are mathematically similar to the ones used in the theory of general relativity: there are four spatial dimensions, (x_1, x_2, x_3, x_4), and we assume the existence of a metric $g_{a,b}(x_1, x_2, x_3, x_4)$ defined over each manifold which describes the curvature of the manifold at the point (x_1, x_2, x_3, x_4). We make the assumption that the metric is everywhere analytic and nowhere singular. (In other words, we assume that g_{ab} is *entire*.)

We will follow GR in the assumption that all physically meaningful quantities can be expressed as local-valued tensor fields T_{ab} that can be calculated from local values of the metric g_{ab}. For example, we assume that the distribution of matter-energy throughout spacetime can be calculated from the stress-energy-momentum tensor. According to this view, we can assume that if two manifolds are not diffeomorphic to one another, then they correspond to different overall physical realities. To illustrate this notion, consider the two 2-dimensional manifolds $\mathcal{M}_1$ and $\mathcal{M}_2$ depicted in Fig. 1 A. The U-shaped curves drawn in the figure represent the curvature of spacetime that is induced by the presence of a particle. In fact, we will adopt the notion that the curvature *is* the particle, a notion that we will explore further in this paper with the 4-geon model of particles as topologically nontrivial structures in the four-dimensional fabric of spacetime. Since the particle in the figure traces out distinct trajectories in $\mathcal{M}_1$ and $\mathcal{M}_2$, we know that $\mathcal{M}_1$ and $\mathcal{M}_2$ represent distinct physical realities. The mathematical expression of this statement is that $\mathcal{M}_1$ and $\mathcal{M}_2$ are not diffeomorphic. A precise definition of these and other related terms (manifold, diffeomorphism, time orientable, bounded, simply connected) is presented for reference in the Appendix, Sec. A.

B. World manifolds and object manifolds

We will use *unbounded* 4-manifolds to represent one possible fine-grained history of an entire "world," and *bounded* three or four dimensional manifolds to represent isolated physical "objects" within a world. These will be called "world" manifolds and "object" manifolds – or simply "objects" – respectively. Conceptually, a *world manifold* will correspond to the physical state and evolution of the entire "universe" from the Big Bang to the Big Crunch (assuming these events actually did take place). World manifolds will be denoted $\mathcal{W}$. Conceptually, an *object manifold* may correspond to the physical state either of a microscopic object (like a particle) or a macroscopic object (like a measuring device or a data collecting device). Object manifolds of dim = 3 will be denoted $\mathcal{O}$; conceptually, we think of these as representing the physical state of an object at a "frozen" instant of time. Object manifolds of dim = 4 will be denoted $\mathcal{Q}$; conceptually, we will think of these as representing the physical state of an object over some range of time from τ_0 to τ_f.

Since we have defined our object manifolds $\mathcal{O}$ and $\mathcal{Q}$ as being bounded, we may give an explicit designation of their boundaries, which we will designate $\mathcal{B}_2$ or $\mathcal{B}_3$. The subscript indicates that these are two or three dimensional surfaces, respectively, which effectively describe the surface of our object manifolds. $\mathcal{B}_2$ and $\mathcal{B}_3$ are themselves manifolds that are oriented to "inside" and "outside" so that the object to which it is the boundary is on the "inside," and the "rest of the universe" is on the "outside." For example, if we wish to consider

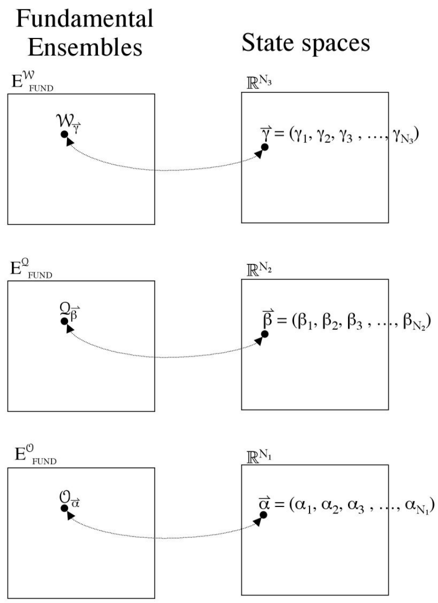

FIG. 2: Ensembles of manifolds and their corresponding state spaces. Left: Ensembles of "all possible" manifolds of given type: $E^{\mathcal{W}}_{FUND}$, $E^{\mathcal{Q}}_{FUND}$, and $E^{\mathcal{O}}_{FUND}$. The elements of these ensembles are $\mathcal{W}_{\vec{\gamma}}$, $\mathcal{Q}_{\vec{\beta}}$, and $\mathcal{O}_{\vec{\alpha}}$. Right: The state spaces $\mathbb{R}^{N_1}$, $\mathbb{R}^{N_2}$, or $\mathbb{R}^{N_3}$ of the ensembles on the left. The elements of these spaces are $\vec{\gamma}$, $\vec{\beta}$, and $\vec{\alpha}$, which are vectors of dimension N_3, N_2, or N_1, respectively. There is a one-to-one mapping between the ensembles on the left and the corresponding state spaces on the right. We assume the existence of a distance measure such that, for example, if two objects $\mathcal{O}_{\vec{\alpha}'}$ and $\mathcal{O}_{\vec{\alpha}''}$ are "almost" the same physical object, then $\vec{\alpha}'$ and $\vec{\alpha}''$ are "really close together" in state space.

the representation of one potential physical state of my laptop computer (Fig. 1 B), we will use an object manifold $\mathcal{O}$ whose two dimensional boundary $\mathcal{B}_2$ describes the surface of a box of size and shape 3cm x 21cm x 27cm. (Note that distance and time intervals are in general understood to be measured from the frame of reference of the object itself.)

C. Ensembles and state spaces

We will use the letter E to denote an *ensemble* (just a fancy word for a *set*) of manifolds, with a superscript denoting the type of manifolds in the ensemble, e.g.: $E^{\mathcal{O}}$, $E^{\mathcal{Q}}$, $E^{\mathcal{W}}$. The ensembles of *all possible* $\mathcal{O}_{\vec{\alpha}}$, $\mathcal{Q}_{\vec{\beta}}$, or $\mathcal{W}_{\vec{\gamma}}$ will be termed the "fundamental" ensembles and labelled with the subscript $FUND$, that is: $E^{\mathcal{O}}_{FUND}$, $E^{\mathcal{Q}}_{FUND}$, or $E^{\mathcal{W}}_{FUND}$. (See Fig. 2.) We define these ensembles such that they contain a *single* copy of each possible manifold; that is, no two manifolds in any of the fundamental ensembles are diffeomorphic to one another. (For reference, Fig. 17 contains a summary of the various spaces defined throughout this paper.)

The subscripts $\vec{\alpha}$, $\vec{\beta}$, and $\vec{\gamma}$ to individual manifolds are parameters that we use to label the individual elements of $E^{\mathcal{O}}_{FUND}$, $E^{\mathcal{Q}}_{FUND}$, or $E^{\mathcal{W}}_{FUND}$, respectively. We conjecture

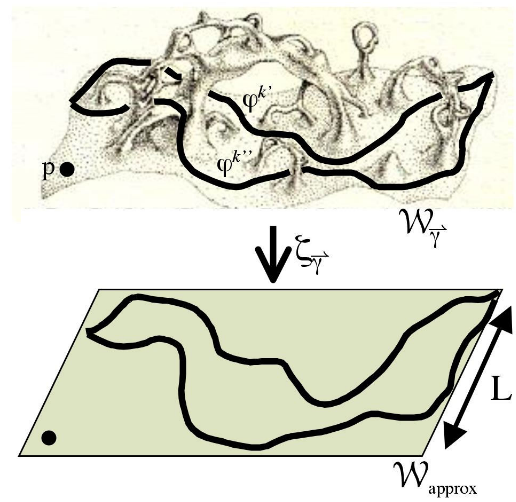

FIG. 3: Top: A two dimensional hyperslice through a four dimensional manifold $\mathcal{W}_{\vec{\gamma}}$ illustrating non simple-connectivity, which has been proposed as a model of quantum foam. We assume that these fluctuations in the curvature of spacetime exist (mostly) on a scale below the Planck-Wheeler length $L = \sqrt{\frac{G\hbar}{c^3}}$ [5]. (Adapted from [21], Fig. 14.3 (c).) Bottom: For any such manifold $\mathcal{W}_{\vec{\gamma}}$ there exists a mapping $\zeta_{\vec{\gamma}}$ that takes points p or curves φ in $\mathcal{W}_{\vec{\gamma}}$ onto an approximately flat, simply connected manifold $\mathcal{W}_{approx}$. In effect, the small scale fluctuations of $\mathcal{W}_{\vec{\gamma}}$ are "smoothed out" by the mapping to $\mathcal{W}_{approx}$.

that it is possible to define parameters $\vec{\alpha}$, $\vec{\beta}$, and $\vec{\gamma}$ as real-valued vectors, i.e. as elements of the vector spaces $\mathbb{R}^{N_1}$, $\mathbb{R}^{N_2}$, or $\mathbb{R}^{N_3}$ respectively, as illustrated in Fig. 2. ($\mathbb{R}$ denotes the real numbers, and N_1, N_2, and N_3 are positive integers.) We define $\mathbb{R}^{N_1}$, $\mathbb{R}^{N_2}$, and $\mathbb{R}^{N_3}$ the "state spaces" of the fundamental ensembles; we will call these $\mathcal{O}$-State Space, $\mathcal{Q}$-State Space, and $\mathcal{W}$-State Space. We will address the question of the dimensionality of $\mathcal{O}$-State Space (i.e., the value of N_1) in a companion paper [20], where we will argue that $\vec{\alpha}$ may be expressed as an infinite-dimensional vector (so $N_1 = \infty$).

D. $\mathcal{W}_{\vec{\gamma}}$ are multiply connected; Spacetime Foam

For several reasons, we will allow our 4-manifolds $\mathcal{W}_{\vec{\gamma}}$ to be multiply connected. One reason is that it will allow us to adopt a model of quantum foam as microscopic fluctuations in the topology of spacetime, as proposed by Wheeler [25], developed by Hawking [13], Thorne [21] and others, and illustrated in Fig. 3. (See also the eloquent expression [24] of this concept by H. Weyl.) We note in passing that it is very difficult to envision what a multiply-connected four-manifold "looks" like. This is why our illustration in Fig. 3 depicts a multiply-connected two-dimensional manifold. Of course, in this type of illustration, what we are doing is to embed the two-dimensional manifold into a three-dimensional space. However, it is important to make note of Gauss' *Theorema*

Egregium, which states that it is always possible to construct an n-dimensional manifold *without* embedding in a higher-dimensional space. Therefore, the multiply connected two-manifold in Fig. 3 does *not* necessitate that we postulate any sort of three-dimensional space in which we embed the two-manifold. Likewise, the fact that the RHF makes use of multiply connected four-manifolds $\mathcal{W}_{\vec{\gamma}}$ does *not* mean that we must either implicitly or explicitly assume the existence of more than four spacetime dimensions. Indeed, we regard as one of the interpretational advantages of the RHF the fact that it does *not* postulate the existence of any extra "unphysical" spacetime dimensions.

In accordance with most of these models, and also in accordance with Hawking's chronology protection conjecture [14] (discussed below; see Sec. II K), we will adopt the assumption that these fluctuations occur "mostly" at a scale below some cutoff length L, and we assume that above this scale, spacetime is "mostly" flat. We justify the postulate that spacetime is mostly flat at the large scale by the simple *observation* that we do not see large-scale topological deformities, such as wormholes, in our everyday experience.

1. $\mathcal{W}_{approx}$

The notion that our four-manifolds $\mathcal{W}_{\vec{\gamma}}$ are flat on a scale above the Planck-Wheeler length L allows us to define a mapping $\zeta_{\vec{\gamma}}$ whose function is to "smooth out" the small scale fluctuations of $\mathcal{W}_{\vec{\gamma}}$. Such a mapping function would take points in $\mathcal{W}_{\vec{\gamma}}$ and map them onto a flat "approximation" four-manifold, $\mathcal{W}_{approx}$, as depicted in Fig. 3.

E. The world ensemble relative to an object manifold

Given any individual object $\mathcal{O}_{\vec{\alpha}}$, we define the world ensemble $E^{\mathcal{W}}_{\vec{\alpha}}$ *relative to* $\mathcal{O}_{\vec{\alpha}}$ as the unique ensemble of all world manifolds $\mathcal{W}_{\vec{\gamma}}$ that satisfy the *boundary condition* defined by $\mathcal{O}_{\vec{\alpha}}$. Another way of stating this would be to say that there exists at least one submanifold of $\mathcal{W}_{\vec{\gamma}}$ that is diffeomorphic to $\mathcal{O}_{\vec{\alpha}}$. In shorthand terminology, we might say that $\mathcal{O}_{\vec{\alpha}}$ is "contained" within $\mathcal{W}_{\vec{\gamma}}$, or that $\mathcal{O}_{\vec{\alpha}}$ "exists inside" $\mathcal{W}_{\vec{\gamma}}$. The notion of an object manifold "existing inside" a world manifold is illustrated in Fig. 4 A. We will refer to $E^{\mathcal{W}}_{\vec{\alpha}}$ as a "relative world ensemble," meaning that it is the ensemble of world manifolds that is defined relative to the object manifold $\mathcal{O}_{\vec{\alpha}}$ (Fig. 4 B).

We will explicitly define the ensemble $E^{\mathcal{W}}_{\vec{\alpha}}$ in such a manner that it may contain more than one copy of any given $\mathcal{W}_{\vec{\gamma}}$. Basically, for each individual occurrence of $\mathcal{O}_{\vec{\alpha}}$ within $\mathcal{W}_{\vec{\gamma}}$, there will be a separate copy of $\mathcal{W}_{\vec{\gamma}}$ within $E^{\mathcal{W}}_{\vec{\alpha}}$. This is one essential difference between a relative world ensemble $E^{\mathcal{W}}_{\vec{\alpha}}$ and the "fundamental" world ensemble $E^{\mathcal{W}}_{FUND}$, which we stipulate has only *one* copy of any given $\mathcal{W}_{\vec{\gamma}}$ in it.

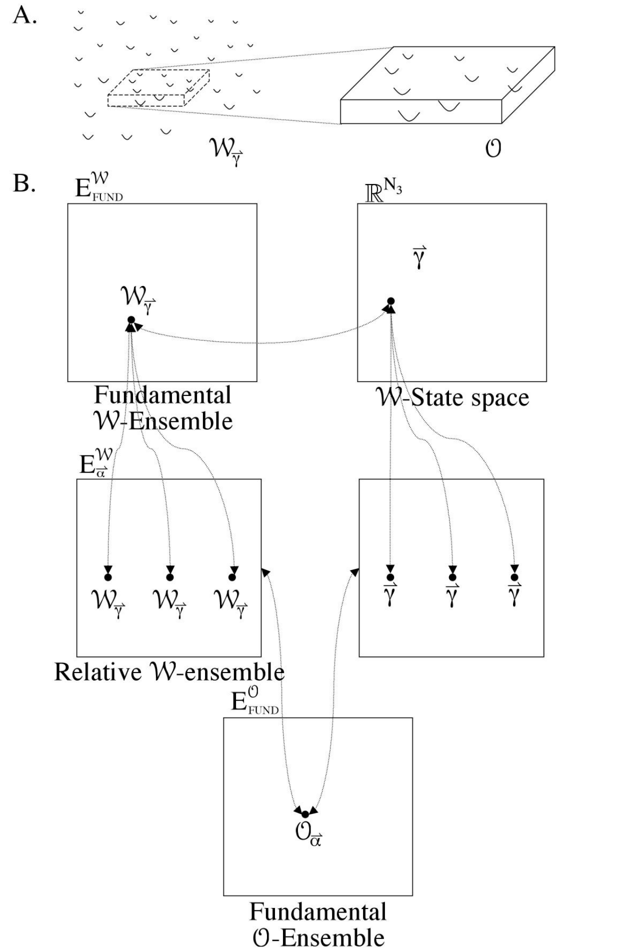

FIG. 4: A. The object manifold representation of my computer $\mathcal{O}_{\vec{\alpha}}$ is diffeomorphic to a submanifold of $\mathcal{W}_{\vec{\gamma}}$. (It is "contained" in $\mathcal{W}_{\vec{\gamma}}$.) Therefore, $\mathcal{W}_{\vec{\gamma}}$ will be an element of $E^{\mathcal{W}}_{\vec{\alpha}}$. B. Any given $\mathcal{O}_{\vec{\alpha}}$ defines an ensemble $E^{\mathcal{W}}_{\vec{\alpha}}$, which we call the "world ensemble relative to $\mathcal{O}_{\vec{\alpha}}$." If there are N copies of $\mathcal{O}_{\vec{\alpha}}$ in $\mathcal{W}_{\vec{\gamma}}$, then there are N copies of $\mathcal{W}_{\vec{\gamma}}$ in $E^{\mathcal{W}}_{\vec{\alpha}}$. The ensemble of $\mathcal{W}_{\vec{\gamma}}$'s (left, middle) could equivalently be expressed as a set of $\vec{\gamma}$'s (right, middle).

F. Any $\mathcal{O}_{\vec{\alpha}}$ may play the role of the "observer"

The formalism of the RHF will place the "observer" at center stage by defining our ensemble of world manifolds – which plays the role of the wavefunction of the universe – explicitly in terms of the mathematical description of the observer itself. We consider this explicit conceptualization of the observer to be one of the major interpretational advantages of the RHF.

Within the framework of the standard formulation of QM, if one were required to specify the "observer" of any given quantum mechanical experiment, one might perhaps identify a data recording device such as a computer that records all essential aspects of the experiment at hand. If one were so inclined, one might even place the brain of the scientist in this role. Alternatively, one could conceivably envision a microscopic object playing the role of the observer. All of these, we note, are *physical entities* of one sort or another. There

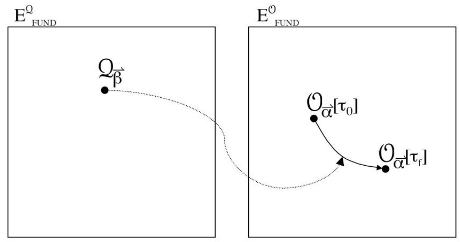

FIG. 5: Any given four dimensional object $\mathcal{Q}$ may be used to form a curve in $E^{\mathcal{O}}_{FUND}$ of three dimensional objects $\mathcal{O}[\tau]$ by making "time slices" through $\mathcal{Q}$. (Any given object in $E^{\mathcal{Q}}_{FUND}$ maps to a curve in $E^{\mathcal{O}}_{FUND}$; however, it is not necessarily the case that all curves in $E^{\mathcal{O}}_{FUND}$ map to a point in $E^{\mathcal{Q}}_{FUND}$.)

are few, if any, guidelines that standard formulations of QM place upon the nature of these entities. It is not even clear that the "observer" should in fact correspond to a physical entity! Within the framework of the RHF, however, we will begin with the explicit designation of the "observer" as a physical entity of some sort. We will, in fact, allow *any arbitrary physical entity* to play the role of the "observer." We will proceed to identify the state of this quantum mechanical "observer" at the beginning of a quantum mechanical experiment with a three dimensional bounded manifold $\mathcal{O}_o = \mathcal{O}_{\vec{\alpha}}$ (the subscript means "observer") that describes the physical state of the observer at the beginning of an experiment. In principle, *any* $\mathcal{O}_{\vec{\alpha}}$ can play the role of the "observer." However, for conceptual purposes, we will – at least at first – generally make use of an $\mathcal{O}_{\vec{\alpha}}$ that corresponds to the physical state of some classical macroscopic recording device, such as a computer, that makes internal records of observations of the external environment in some classical way, e.g. in the form of bits that correlate with external measurements. In this sense, we will borrow one of the central tenets of Everett's relative state formulation [2]; namely, we require that all meaningful observations made by the observer of the *external environment* be manifest in some fashion, e.g. in the form of electronic bits, upon the *internal structure* of the observer itself. In a companion paper [20], several mathematical constraints are proposed to be applied to the choice of $\mathcal{O}_o = \mathcal{O}_{\vec{\alpha}}$ so that we may make the continued elaboration of the formalism of the RHF more tractable.

G. "Time slice" through $\mathcal{Q}$ and the "temporal support" of an object

We will symbolize the "time slice" through a four-dimensional object $\mathcal{Q}$ at $\tau \in [\tau_0, \tau_f]$ using the notation: $\mathcal{O}[\tau]$, that is, as a three-dimensional object. For our purposes, we will restrict the boundaries $\mathcal{B}_3$ of our 4-objects $\mathcal{Q}$ in such a way that for any two $\tau_1, \tau_2 \in [\tau_0, \tau_f]$, the boundary of $\mathcal{O}[\tau_1]$ is the same as the boundary of $\mathcal{O}[\tau_2]$. In other words, we define our 4-objects such that the "size and shape" of $\mathcal{Q}$ is constant over time, and we may assert that the set of all "time slices" through any $\mathcal{Q}$ defines a continuous curve in $E^{\mathcal{O}}_{FUND}$, as depicted in Fig. 5. Thus, any single $\mathcal{Q}$ can be thought of as one possible time dependent evolution of a three dimensional object.

In the simplest case, $\mathcal{B}_2$ has the topology of a simply connected, two-dimensional sphere (that is, $\mathcal{B}_2 = (S^2, g_{ab}(x_1, x_2, x_3, x_4))$), and $\mathcal{B}_3$ has the topology $S^2 \times \mathbb{R}$. $\mathcal{B}_2$ and $\mathcal{B}_3$ are oriented boundaries, with the object being on the inside and the rest of the universe being on the outside. We will typically consider $\mathbb{R}$ in the expression $S^2 \times \mathbb{R}$ to correspond to the time dimension. Recall that our world manifolds $\mathcal{W}$ are not necessarily time-orientable. In other words, if we identify one of the four coordinates at *one specific* point (x_1, x_2, x_3, x_4) in $\mathcal{W}$ as the time coordinate, then there is *not* necessarily any meaningful way that we can extrapolate this specification in a consistent fashion that allows us to say which coordinate is the time coordinate at *any* point in $\mathcal{W}$. However, we will (typically) assume that our object manifolds $\mathcal{Q}$ and $\mathcal{O}$ *are* time orientable, and that, as stated above, for an object manifold $\mathcal{Q}$ whose boundary has the topology $S^2 \times \mathbb{R}$, the time coordinate is represented by $\mathbb{R}$. This specification of the time coordinate will be referred to as the *temporal support* of $\mathcal{Q}$ or $\mathcal{O}$. The concept of temporal support will be discussed further in [20].

H. Tree diagram and probability measure

One possible fine-grained evolution of the observer $\mathcal{O}_o = \mathcal{O}_{\vec{\alpha}}$ over time may be represented mathematically as a four-dimensional bounded manifold $\mathcal{Q}_{\vec{\beta}}$, where $\mathcal{O}_{\vec{\alpha}}$ is diffeomorphic to the time slice of $\mathcal{Q}_{\vec{\beta}}$ at $\tau = \tau_0$. There will in general be many such manifolds $\mathcal{Q}_{\vec{\beta}}$, each of which defines a curve in $E^{\mathcal{O}}_{FUND}$ (Fig. 6, A). The set of all such $\mathcal{Q}_{\vec{\beta}}$ defines a branching tree-like structure in $E^{\mathcal{O}}_{FUND}$ (Fig. 6, A). This branching tree-like structure can be depicted in slightly different manner as shown in Fig. 6, B, by plotting the variable $\vec{\alpha}$ versus the proper time parameter τ. We call this the the "tree diagram" of the observer. We make the assumption that the evolution of the observer $\mathcal{O}_{\vec{\alpha}}$ over time is a "random walk" progression through $E^{\mathcal{O}}_{FUND}$ (or down the tree diagram) that is restricted to follow one of the trajectories of the tree diagram. According to this **random walk hypothesis**, each point along the tree diagram is associated with a real-valued nonnegative number in the range $[0, 1]$ that we call the "probability measure." Given an observer $\mathcal{O}_{\vec{\alpha}}$ and its tree diagram, we assign the probability measure of the tree diagram at the origin of the tree to be equal to 1. For each point of the tree diagram distal to the origin, the probability measure can be calculated through the repeated application of this rule: *Given a branch point with N discrete branches, the probability measure of each distal branch equals the probability measure of the proximal branch multiplied by 1/N.* This concept is illustrated in Fig. 7 A.

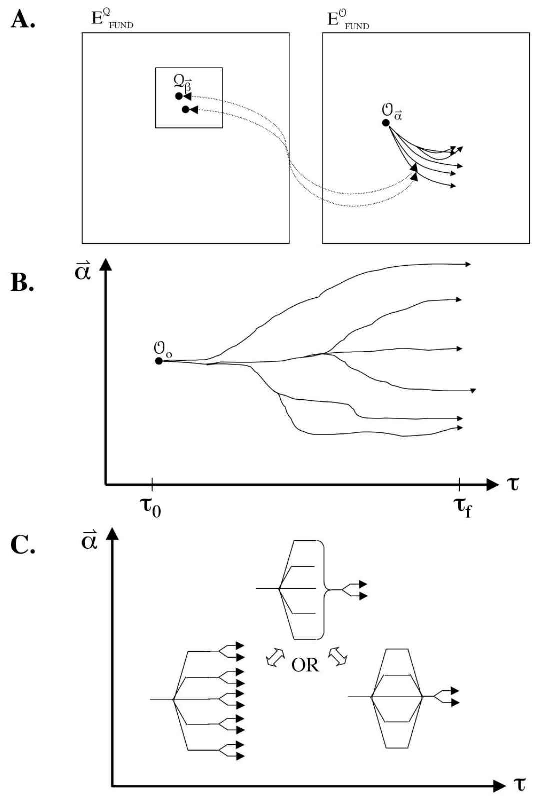

FIG. 6: **The tree diagram of an observer $\mathcal{O}_{\vec{\alpha}}$. A.** Given a choice of our observer $\mathcal{O}_{\vec{\alpha}}$, and time interval $T = \tau_f - \tau_0$, we draw every curve in $E^{\mathcal{O}}_{FUND}$ that begins at $\mathcal{O}_{\vec{\alpha}}$ and that corresponds to a point in $E^{\mathcal{Q}}_{FUND}$. **B.** The tree diagram of $\mathcal{O}_{\vec{\alpha}}$, which essentially shows the same information as in A; the curves are simply depicted in a different format ($\vec{\alpha}$ versus τ) to emphasize the time dependence. Note that $\vec{\alpha}$ is multidimensional, even though we have drawn it as one-dimensional. For any given $\mathcal{O}_{\vec{\alpha}}$ and $T = \tau_f - \tau_0$, there is one and only one tree diagram. **C. Notation:** The tree diagram in the middle may be used for notational convenience to represent either of the tree diagrams on the left or on the right. The one on the left branches in many directions, with each distal branch having a (mostly) identical appearance. Even though each of these distal branches is in a different location in state space, we assume that each of these distal branches are "similar enough" that they can each be represented by a single (distal) tree. The one on the right branches and then fuses.

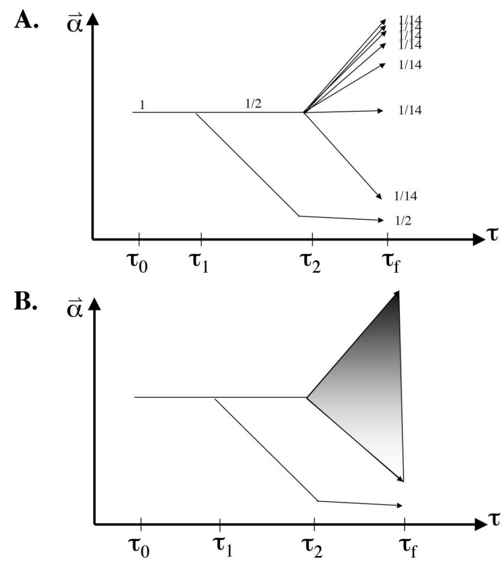

FIG. 7: The random walk hypothesis can be used to calculate the probability measure at any point along the tree diagram. A: Depiction of a tree diagram made out of discrete curves. At time τ_1, the tree branches into two branches. At time τ_2, the upper branch splits further into seven further branches. Each point along the diagram is associated with a "probability measure;" the probability measure of several points along the diagram are shown. *B:* The same tree diagram in A except that we have taken the number of branches at τ_2 to infinity, so that the branching at time τ_2 has become continuous. In this case, the probability of following either branch at τ_1 remains 1/2, while the probability of following any trajectory past τ_2 (along the upper branch at τ_2) may be conceptualized as the flow of a current through state space, with the "current density" depicted (roughly) by the shading in the diagram.

1. *Approximation methods*

For our statement of the random walk hypothesis, we have implicitly assumed that our tree diagrams will branch in a discrete manner (as illustrated in Fig. 7 A). If the number of branches at a given branch point is N, then in the limit as N approaches infinity, (as illustrated in Fig. 7 B), we obtain what may be viewed more appropriately as a continuous "flow" or "current" through state space. Throughout the development of the RHF, there will be several situations in which we will assume the existence of a mathematical method of approximation such as this one whereby a continuous process may be approximated as a discrete one, with the continuous process existing in the limit as some parameter is allowed to approach infinity. This situation is not unlike the basic method of the FPI, in which the set of "all possible paths" may be approximated by a set of discrete "representative" paths (as outlined in the Appendix).

(The notion of presenting all tree diagrams as having a discrete structure may make Regge calculus particularly suitable for the RHF. In this, one decomposes the spacetime manifold into a simplicial complex.)

I. Statements and observables

Suppose we wish to evaluate a statement or a proposition X such as the following: "There exists an experimental apparatus that is set up in such-and-such a configuration." How do we translate an English-language sentence such as this into the terminology of the RHF? Our first goal will be to evaluate whether the statement X is true or false, which we will do *relative to* a world manifold $\mathcal{W}_{\vec{\gamma}}$.

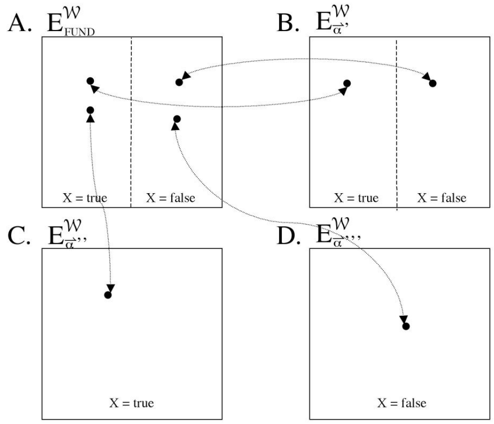

FIG. 8: Evaluation of a statement X. Any ensemble of world manifolds (A, the fundamental ensemble; as well as B - D, relative ensembles) can be divided into the subset of $\mathcal{W}_{\vec{\gamma}}$ such that statement X is true and the subset of $\mathcal{W}_{\vec{\gamma}}$ such that X is false. Relative to the object manifold $\mathcal{O}_{\vec{\alpha}''}$, the statement X is true; relative to $\mathcal{O}_{\vec{\alpha}'''}$, the statement X is false; and relative to $\mathcal{O}_{\vec{\alpha}'}$, the statement X is in a "superposition" of true and false.

1. *Definition of a statement X*

Essentially, the definition of our statement X will be in terms of an object manifold $\mathcal{O}^X$ that matches the description of our experimental apparatus in the specified configuration. ($\mathcal{O}^X$, we note in passing, is macroscopic.) Of course, any *individual* object manifold $\mathcal{O}^X$ must provide a description of our experimental apparatus down to the tiniest detail, including the exact positions and momenta of every particle. Typically, therefore, there will be an ensemble $E^{\mathcal{O}}_X$ of 3-manifolds that will match the general description of our experimental apparatus in its required configuration.

2. *Evaluation of a statement relative to a world manifold $\mathcal{W}_{\vec{\gamma}}$*

Our next step will be to determine whether the object manifold $\mathcal{O}^X$ does or does not exist within the world manifold $\mathcal{W}_{\vec{\gamma}}$. We will make use of the concept of an object manifold being "contained" within a world manifold that we presented earlier in Sec. II E. Basically, the statement X is true relative to $\mathcal{W}_{\vec{\gamma}}$ iff $\mathcal{O}^X$ is contained within $\mathcal{W}_{\vec{\gamma}}$. Otherwise, it is false.

If we define our statement X in terms of an *ensemble* $E^{\mathcal{O}}_X$ of object manifolds (as will usually be the case), then X is considered to be false with respect to $\mathcal{W}_{\vec{\gamma}}$ iff *none* of the $\mathcal{O}^X$ in $E^{\mathcal{O}}_X$ are contained within $\mathcal{W}_{\vec{\gamma}}$, and true iff one (or more) of the $\mathcal{O}^X$ are contained within $\mathcal{W}_{\vec{\gamma}}$.

In this way, any ensemble of world manifolds may be expressed as the union of disjoint subsets such that X is true and X is false, as depicted in Fig. 8.

3. *Evaluation of a statement relative to the observer $\mathcal{O}_o = \mathcal{O}_{\vec{\alpha}}$*

We are now in a position to evaluate a statement X with respect to an observer $\mathcal{O}_o = \mathcal{O}_{\vec{\alpha}}$ and its associated world ensemble $E^{\mathcal{W}}_o = E^{\mathcal{W}}_{\vec{\alpha}}$. Let us define our statement X as we did above. This will allow us to go through each world manifold $\mathcal{W}_{\vec{\gamma}}$ in $E^{\mathcal{W}}_o$, one at a time so to speak, and determine whether X is true or false relative to that particular $\mathcal{W}_{\vec{\gamma}}$. So from the overall perspective of our observer $\mathcal{O}_o$, any given statement X is considered to be true (T) if X is true for every manifold in $E^{\mathcal{W}}_o$ (Fig. 8 C), false (F) for $\mathcal{O}_o$ if it is false for every manifold in $E^{\mathcal{W}}_o$ (Fig. 8 D), and in a "superposition" (S) of true/false if it is true for some manifolds in $E^{\mathcal{W}}_o$ and false in others (Fig. 8 B). Since any given point on a tree diagram corresponds to an individual object $\mathcal{O}_{\vec{\alpha}}$, then at *any* point along the tree diagram, we can assign a value T, F, or S to a statement X (Fig. 9 A).

A true-false statement is one of the simplest types of statements. There are many ways to build more complex types of statements. One useful type of statement (which we will make use of later in this paper) is an *array* of statements. We may build up a set or array of statements X_b, with b an integer in $[0, B]$, using (for example) the statement: X_b = "We have set up an experiment using such-and-such a configuration, and out of B detectors, the particle is detected by detector b." (The value b = 0 indicates absorption of the particle by a barrier.) Let us assume that if our experiment is properly set up (which we assume it is for every manifold in $E^{\mathcal{W}}_o$), then the particle is either absorbed by a barrier or is detected by one and only one detector. Thus, the array of statements X_b divides the ensemble $E^{\mathcal{W}}_o$ into disjoint subsets, as depicted in Fig. 9 B (top). For any given point $\mathcal{O}_{\vec{\alpha}}$ on the tree diagram, we can define the set [B] as the list of b values such that X_b is true for *at least one* $\mathcal{W}_{\vec{\gamma}}$ in $E^{\mathcal{W}}_{\vec{\alpha}}$ (Fig. 9 B, bottom). At any point along the tree diagram, the set [B] will always have at least one b-value in it; that is, [B] will never be the empty set. If [B] contains more than one b-value at a point $\mathcal{O}$ on the diagram, then we may say (speaking very loosely) that the particle is in a "superposition" of each of these b-values at that point on the diagram. If [B] contains only one b-value at a point $\mathcal{O}$ on the diagram, then we may say (again, speaking loosely) that the particle has "collapsed" onto that b-value. See the legend to Fig. 9 for more details regarding such slang (but very useful nonetheless) terminology.

Also note that Fig. 9 illustrates the general principle that for $\vec{\alpha}''$ distal to $\vec{\alpha}'$, $E^{\mathcal{W}}_{\vec{\alpha}''}$ is necessarily a *subset* of $E^{\mathcal{W}}_{\vec{\alpha}'}$. This general principle will be discussed in further detail in a companion paper [20].

4. *A* statement *in the RHF is analagous to an* observable *in the standard formulation of QM*

Our definition of a statement is sufficiently generalized such that just about any English-language sentence can be cast in the form of a statement X. Through a suitable choice of words, therefore, a statement X can be constructed to play the role of the quantum mechanical operator/observable. For example: consider the most simple and basic of operators in the

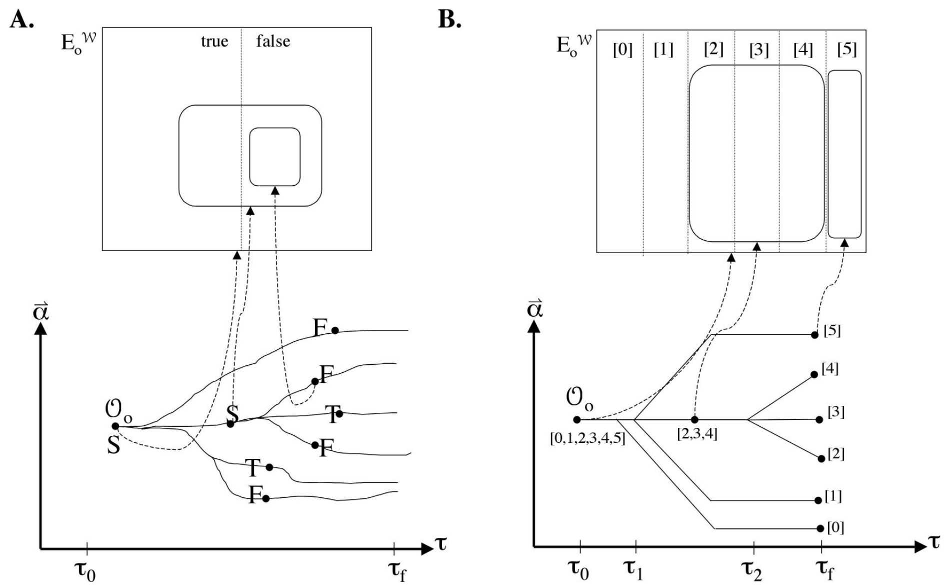

FIG. 9: **Statements and the tree diagram. A. A true/false statement.** Above: The ensemble $E^{\mathcal{W}}$ is divided into two subsets, those for which a statement X is true and those for which it is false. Below: A point $\mathcal{O}_o = \mathcal{O}_{\vec{\alpha}}$ in the tree diagram maps to an ensemble in history space. The statement is shown to be true (T), false (F), or in a superposition (S) of true/false at several different points depending on how it maps into history space. **B. A multivalued statement.** Top: The array of statements X_b divides the ensemble $E^{\mathcal{W}}$ into six disjoint subsets, with X_b defined as the reading on the dial of a measurement apparatus that measures the position of a particle, and with b taking values $b \in [0, 1, 2, 3, 4, 5]$. Below: With respect to an observer $\mathcal{O}_o[\tau]$, the dial of the measurement apparatus is said to be in a superposition iff $\mathcal{O}_o[\tau]$ maps to an ensemble E in state space that overlaps with more than one of the six subsets. In slang terminology, we might say that at time τ_2, the particle under observation "collapses" from a superposition of states 2, 3, and 4 onto just one of the individual states 2, 3, or 4. We might also say that the particle "decoheres" from a superposition of states 2, 3, 4 onto one of the states 2, 3, or 4, or that the observer "makes a measurement" of the position of the particle. Although we are using here the example of a particle in configuration space, we can use the same basic ideas to depict the "collapse" of any observable.

standard formulation, the position operator. In the framework of the RHF, we simply start with the sentence: "There exists a particle that is located within the particle detector at position b_i" and proceed as we just described in the analysis of this statement.

By way of illustration, consider Fig 9 A as a representation of the Schrödinger's Cat experiment. In this case, we imagine that $\mathcal{O}_o$ defines an ensemble $E_o^{\mathcal{W}}$ that corresponds to the initial setup of a typical Schrödinger's Cat experiment, and the statement X is defined as "the cat is alive." Thus X = true means that the cat is alive, and X = false means that the cat is dead. In Fig 9 A, each point on the tree diagram can be labelled as T, F, or S as discussed above. We have depicted several branch points on the tree diagram that correspond to a transition from S to distal branches of T for some branches (meaning that the cat is alive) and F for others (meaning that the cat is dead). Such an occurrence corresponds to the moment of observation of the state of the cat. We would say that this point on the tree diagram corresponds to the instant of "collapse." (Note that collapse, like so many other things in the RHF, is explicitly defined *with respect to the observer*.)

J. Relative probability

We can use the concepts of probability measure of a tree diagram (Fig. 7) and an array X_b of statements to determine the overall probability relative to $\mathcal{O}_o = \mathcal{O}_{\vec{\alpha}}$ that a particle will ultimately be detected by a detector b out of a total of B distinct detector elements. As depicted in Fig. 10, this is calculated by determining the sum of the probability measure over *every branch* for which the tree diagram has collapsed onto that particular b-value. (Fig. 10). For example, if there are two branches corresponding to collapse onto the particle being at detector $b = 2$, then the total probability of finding the particle at detector 2 is the sum of the probability measure of these two branches. In the example of Fig. 10, this gives us $1/4 + 1/8 = 3/8$, that is, a 37.5 percent chance that the particle will be detected by detector 2.

K. Hawking's Chronology Protection Conjecture (CPC)

It is well known that the allowance of wormholes in the fabric of spacetime introduces difficulties with our intuitive notions of causality. Certain types of multiple connectedness, for example, may allow for the existence of a "time machine."

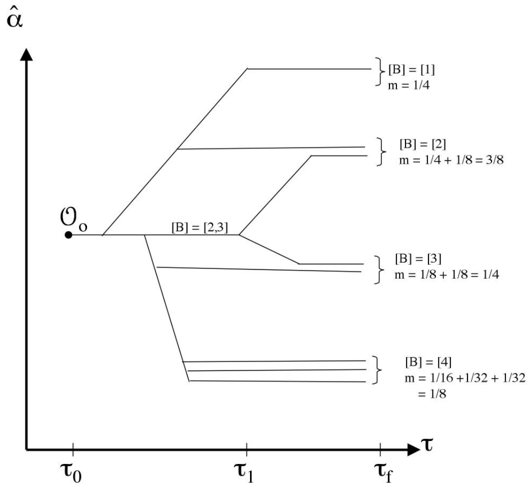

FIG. 10: Relative probability. A hypothetical tree diagram is shown. We wish to calculate the probability relative to $\mathcal{O}_o = \mathcal{O}_{\vec{\alpha}}$ that a particle will eventually be detected by detector b. To do this, we follow the tree along each branch until [B] contains one and only one b-value, we calculate the measure at each of these points, and we add up the measure of each branch for which [B] = [b].

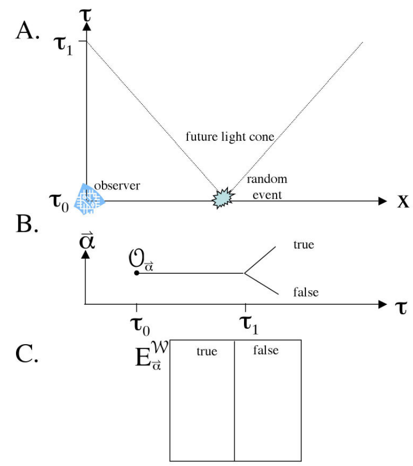

FIG. 11: The Chronology Protection Conjecture.

The difficulty with causality posed by the existence of time machines is illustrated through the famous "grandfather paradox," in which a man travels backwards in time through a wormhole and kills his grandfather, thus (paradoxically) preventing his own birth.

These considerations have prompted Stephen Hawking to pose a "chronology protection conjecture" [14] (CPC). Essentially, Hawking's CPC states that Nature forbids the existence of "time machines." However, as discussed by Li and Gott [15], Hawking's chronology protection conjecture does not apply to quantum time machines that are confined to small spacetime regions. In the context of the RHF, we will assume that Hawking's CPC applies at the large scale, but not at the small scale. That is, "most" manifolds $\mathcal{W}_{\vec{\gamma}}$ "look like" the one depicted in Fig. 3 A. Of course, when we say "most," what we mean is that manifolds with large wormholes (above some cutoff length) are *less probable* than those without large wormholes. We present the CPC as being similar to the Second Law of Thermodynamics in the sense that both are "usually" true without being necessarily "always" true. To put it another way, the CPC and the Second Law both make statements about what is most probable; in both cases, there is always a nonzero probability that they may be violated.

1. *The CPC and Tree Diagrams*

The CPC has important implications for how an observable or statement can split a tree diagram into branches. In particular, consider an observable X that corresponds to the outcome of an event at some particular spacetime location. According to the CPC, the form of a tree diagram can be a function of the outcome of this event if and only if this event is within the past light cone of the observer.

Consider, for example, Fig. 11. The blue star represents a random event that results in an object being in one of two states that we label "true" and "false" and which we assume *not* to be entangled with any object that is not in its immediate vicinity. For example, the random event might be the creation of a particle in the spin up (true) or spin down (false) state. We also assume that the state of the particle is "leaked" into the environment so that it is observable in the classical sense. According to the CPC, the observer will be unable to observe the outcome of this event until a time T after the event happens, where T equals the distance X between the observer and the random event divided by the speed of light c, $T = X/c = \tau_1 - \tau_0$. (with values measured in the frame of reference of the observer). This is reflected in the tree diagram by the fact that the diagram remains in superposition of T and F states until the proper time τ_1. According to the CPC, the structure of the tree diagram may *not be a function of the state of the particle at any time prior to* τ_1.

We have been very careful not to violate the CPC at any point in our development of the RHF. In a Part II [19], for example, we perform an analysis of the EPR experiment, taking great care that the CPC is not violated in our analysis. In our demonstration that the RHF and the FPI make the same predictions (below), we will once again take great care that the CPC is not violated.

L. Path approximation in $\mathcal{W}_{approx}$

The notion of spacetime foam will allow us to assume the existence of a mapping $\zeta_{\vec{\gamma}}$ that takes points in a non simply connected manifold $\mathcal{W}_{\vec{\gamma}}$ into a mostly flat, simply connected approximation manifold $\mathcal{W}_{approx}$, as illustrated in Fig. 3.

With respect to any given observer $\mathcal{O}_{\vec{\alpha}}$ and its associated world ensemble $E^{\mathcal{W}}_{\vec{\alpha}}$, we may define an approximation manifold $\mathcal{W}_{approx}$ as discussed earlier and depicted in Fig. 3. Note that with respect to any given observer, we have a *single* $\mathcal{W}_{approx}$, but an *ensemble* of $\mathcal{W}_{\vec{\gamma}}$. As stated earlier, we assume the existing of mappings $\zeta_{\vec{\gamma}}$ that take points p or curves φ in $\mathcal{W}_{\vec{\gamma}}$ onto $\mathcal{W}_{approx}$. Let us see how $\mathcal{W}_{approx}$ may be of some use.

Consider the *equivalence classes* (as defined in the Appendix) of the set of all paths connecting any two points in a manifold $\mathcal{W}_{\vec{\gamma}}$. Each equivalence class will have what is called a "class representative" path φ, which is the "shortest" path among all the paths in the equivalence class. As depicted in Fig. 3, $\varphi^{k'}$ and $\varphi^{k''}$ can be seen to belong to distinct equivalence classes because the one cannot be smoothly deformed into the other; in effect, the small scale wormholes depicted in the figure constitute barriers to the deformation of $\varphi^{k'}$ into $\varphi^{k''}$. For the remainder of this paper, whenever we mention a path φ in a world manifold $\mathcal{W}_{\vec{\gamma}}$, it is to be understood (unless stated otherwise) that φ is one of these *class representative* paths.

Within $\mathcal{W}_{approx}$, let us define the set of all paths connecting any two points using the standard technique of the Feynman path integral (FPI). (See the Appendix for an overview of the FPI. We will not go into the details of how all paths are enumerated in the FPI technique; this is a complex topic that belongs to the study of the calculus of variations.) We merely wish to propose in this section that there is a one-to-one correspondence between the set of all class representative paths φ in $\mathcal{W}_{\vec{\gamma}}$ as outlined above, and the set of all paths φ in $\mathcal{W}_{approx}$ as enumerated using the standard techniques of the FPI. In other words, we suppose that if you take the set of all class representative paths between two points in $\mathcal{W}_{\vec{\gamma}}$ and map them to the approximation manifold $\mathcal{W}_{approx}$ using the mapping function $\zeta_{\vec{\gamma}}$, then you will end up with the same set (or at least a reasonable approximation) of all paths φ that you would get via the FPI technique (the calculus of variations).

Given a suitable definition for the lagrangian $L_{\vec{\gamma}}(x_1, x_2, x_3, x_4)$ over $\mathcal{W}_{\vec{\gamma}}$, we may define the general relativistic action $S_{gr}(\varphi^k)$ of the path φ^k as the integral of the lagrangian along the length of the path. As previously discussed, each of these paths has its image in $\mathcal{W}_{approx}$. Given the image of the path φ^k in $\mathcal{W}_{approx}$, we can calculate a classical action $S_{cl}(\varphi^k)$ in the manner described by the Feynman path integral formulation, as outlined in the appendix. We will conjecture that the classical action $S_{cl}(\varphi^k)$ as calculated using the FPI methodology is equal in close approximation to the general relativistic action $S_{gr}(\varphi^k)$ as calculated using the method outlined above. We will see this conjecture play an important role in the considerations of Sec. IV.

M. Superposition and collapse

Let us once again explore how the concept of "superposition" is translated into the language of the RHF. Suppose we are presented with an observer $\mathcal{O}_o = \mathcal{O}_{\vec{\alpha}}$ with its associated relative world ensemble, $E^{\mathcal{W}}_{\vec{\alpha}}$. Suppose that $\mathcal{O}_{\vec{\alpha}}$ corresponds to the state of a computer that has an internal representation of an external (external to $\mathcal{O}_{\vec{\alpha}}$) experimental setup, and that this internal representation is an accurate reflection of the external environment. Suppose further that the experimental setup includes a particle that has been prepared in a superposition of states (in the standard quantum mechanical sense of the word) $|1>$ and $|2>$. Let us assume that the general *position* of the particle is in a definite location (relative to the observer), and that $|1>$ and $|2>$ correspond to, say, some other physical attribute; for example, the orientation of its spin axis. In the language of the RHF, the particle is represented by an object $\mathcal{O}^P$ (where the P reflects that it corresponds to a "particle"). Since the particle exists in a superposition of two states, we will in fact require a separate 3-object for each of these two states: $\mathcal{O}^P_1$ and $\mathcal{O}^P_2$. Within any given $\mathcal{W}_{\vec{\gamma}}$ that is an element of $E^{\mathcal{W}}_{\vec{\alpha}}$, there will be an object $\mathcal{O}^P_{\vec{\gamma}}$ that corresponds to the particle in question which is in some definite position relative to the observer. However, in some cases, we will have $\mathcal{O}^P_{\vec{\gamma}} = \mathcal{O}^P_1$, whereas in others, we will have $\mathcal{O}^P_{\vec{\gamma}} = \mathcal{O}^P_2$. World manifolds $\mathcal{W}_{\vec{\gamma}}$ that fit into the first category we will define to be elements of the set $E^{\mathcal{W}}_1$, and world manifolds $\mathcal{W}_{\vec{\gamma}}$ that fit into the second category we will define to be elements of the set $E^{\mathcal{W}}_2$. $E^{\mathcal{W}}_1$ and $E^{\mathcal{W}}_2$, therefore, must be disjoint ensembles, and must be subsets of $E^{\mathcal{W}}_{\vec{\alpha}}$. As long as $E^{\mathcal{W}}_1$ and $E^{\mathcal{W}}_2$ are each nonempty sets, we say that the particle in question exists in a superposition of the states $|1>$ and $|2>$. As our observer $\mathcal{O}_o = \mathcal{O}_{\vec{\alpha}}$ flows through state space, the composition of its associated relative ensemble $E^{\mathcal{W}}_{\vec{\alpha}}$ will evolve. If, at a specific instant of proper τ, one of these subsets (say, $E^{\mathcal{W}}_1$) becomes empty, with $E^{\mathcal{W}}_2$ nonempty, we say that the state of the particle has "collapsed" onto the definite state $|2>$.

So in the overall framework of the RHF, what causes "collapse?" According to the workings of our scheme, collapse occurs at the point in time τ when the observer evolves into a configuration that is mathematically inconsistent with one (or more) of the states of the particle. That is to say that $\mathcal{O}_o$ and, say, $\mathcal{O}^P_2$ cannot coexist within any single given world manifold $\mathcal{W}_{\vec{\gamma}}$. In standard parlance, one sometimes uses the word "decoherence" to describe the process whereby a particle evolves from being in a superposition of states to being in a collapsed state. In the terminology of quantum computing, one might say that a particle that remains in a superpositional state is "shielded" from interaction with its environment, and that the shielding is what prevents the decoherence of the particle into a definite state. Let us examine, then, the process of decoherence from the prespective of the RHF. As an example, suppose we have a particle that exists in a superposition of vibrational states $|1>$ and $|2>$, as for example the ammonia molecule, which has in fact been exploited in the field of quantum computing for the purpose of designing a qubit (ref). In the RHF representation of one of these particles, we can imagine that the vibration of the particle will send "ripples" through the fabric of spacetime in the form of gravitational waves. Tiny though they may be, these ripples will differ between world manifolds $\mathcal{W}_{\vec{\gamma}'}$ and $\mathcal{W}_{\vec{\gamma}''}$, assuming that the vibrational state of the particle differs between $\mathcal{W}_{\vec{\gamma}'}$ and $\mathcal{W}_{\vec{\gamma}''}$. Once (but not before!) these ripples reach the location of the observer $\mathcal{O}_o$, the observer itself will be forced into two vibrational states. This, in turn, corresponds to a branch point on the tree diagram of the observer; if the observer evolves down one branch, then the particle has effectively collapsed onto that vibrational state; if the observer evolves down the other branch, then this corresponds to collapse of the particle onto the other vibrational state. For a particle to remain shielded, therefore, we require that these two vibrational modes of the particle either do not emit gravitational waves, or that they emit *identical* waves. We might say that a particle that emits information in this manner – in the form of gravitational waves of some sort – has "leaked information" into its environment, thus resulting in its decoherence.

As a final note, we mention that in standard quantum mechanics, the descriptors of many states, such as the position operator $|x\ \ >$, are continuous-valued rather than discrete. Therefore, the tendency will be for any given object, whether it be microscopic (like a particle) or macroscopic (like a cat), to exist in a superposition of an *infinite* number of states, requiring the appearance of an *infinite* number of world manifolds $\mathcal{W}_{\vec{\gamma}}$ in the relative ensemble $E^{\mathcal{W}}_{\vec{\alpha}}$. When an object "leaks" information into its environment in the manner discussed above, it is reasonable to ask: what does this information (in the form of gravitational waves) convey to the observer about the state of the object? One particularly relevant question, given the workings of the tree diagram, would be: In what order does this information get leaked? This is a relevant question because the temporal *order* by which an observer makes an observation of the internal state of a particle will have a dramatic effect on the probability of collapse onto these particular states. We will see this notion to have an important role in our demonstration of the equivalence between the RHF and the FPI, later in this paper.

III. THE 4-GEON MODEL OF ELEMENTARY PARTICLES

In this section, we will propose characteristics of the 3- and 4-object manifolds $\mathcal{O}^P$ and $\mathcal{Q}^P$ that we use to model elementary particles. Our choices in this model are guided by the desire that the RHF give rise to quantum statistics. In Sec. IV, we will attempt to show how the properties that we propose in this section serve us in the pursuit of this goal.

Within the RHF, the physical representation of a particle within a world manifold $\mathcal{W}$ over time from its creation at time τ_0 to its destruction at time τ_f is given by a 4-object $\mathcal{Q}^P$ (the superscript indicates "particle"), with $\mathcal{O}[\tau]^P$ representing the physical state of the particle at the point in time τ. (From the perspective of the observer $\mathcal{O}_o = \mathcal{O}_{\vec{\alpha}}$, the particle is more properly expressed as an ensemble of 4-objects $\mathcal{Q}^P$, each individual 4-object $\mathcal{Q}^P$ being contained within a different world manifold $\mathcal{W}_{\vec{\gamma}}$ within the ensemble $E^{\mathcal{W}}_o$.)

The boundaries of $\mathcal{Q}^P$ and $\mathcal{O}[\tau]^P$ are $\mathcal{B}_3$ and $\mathcal{B}[\tau]_2$, which are closed three- and two-dimensional surfaces, respectively. As discussed previously, these surfaces are oriented to indicate "inside" versus "outside." We will begin our characterization of the properties of $\mathcal{Q}^P$ and $\mathcal{O}[\tau]^P$ by adopting Hadley's concept of a 4-geon [7] [8] [9] [10] [11] [12] :

Definition 1. *4-geons. $\mathcal{Q}^P$ and $\mathcal{O}[\tau]^P$ are semi-Riemannian spacetime manifolds, which are solutions of Einstein's equations of general relativity. $\mathcal{Q}^P$ and $\mathcal{O}[\tau]^P$ are topologically non-trivial, with a non-trivial causal structure, and are asymptotically flat (Axiom 2) and particle-like (Axiom 1).*

Implicit in our assumption of a "non-trivial causal structure" of 4-geons is the notion that 4-geons are (or at least can be) multiply connected. We may imagine that different types of particles will be characterized by different types of structures. In this paper, we will not go into the differences between different particle types, but will rather restrict ourselves to a few essential characteristics that we propose as common to *all* particle types. Let us continue our basic characterization with Hadley's particle-like axiom [8]:

Axiom 1. *Particle like. For any given closed, oriented two-dimensional surface $\mathcal{B}_2$, an experiment to determine the presence of a particle will yield a true or a false value only.*

In other words, any given 3-object $\mathcal{O}$ with boundary $\mathcal{B}_2$ either is or is not a "particle," based upon its internal structure.

We now turn to Hadley's axiom of asymptotic flatness [8]:

Axiom 2. *Asymptotic flatness. Outside $\mathcal{Q}^P$ (that is, in the region exterior to $\mathcal{B}_3$), spacetime is topologically trivial and asymptotically flat with an approximately Lorentzian metric.*

The axiom of asymptotic flatness is consistent in spirit with our earlier discussion of the chronology protection conjecture, according to which our manifolds $\mathcal{W}_{\vec{\gamma}}$ are flat on the large scale. We might imagine that any and every possible variety and combination of "non-trivial causal structure" corresponds to a particular particle (or group of particles). Let us assume that for any given world manifold $\mathcal{W}_{\vec{\gamma}}$, we can enumerate all of the 4-geons $\mathcal{Q}^P$. Suppose that we remove each of these $\mathcal{Q}^P$'s from $\mathcal{W}_{\vec{\gamma}}$ and replace each of them with 4-manifolds $\mathcal{Q}$ that have trivial topological structures, by "sewing" them into the region where each of the particles ($\mathcal{Q}^P$)'s used to be. The resulting manifold we will call $\mathcal{W}^e_{\vec{\gamma}}$ (where e indicates "empty of particles"). $\mathcal{W}^e_{\vec{\gamma}}$, we may imagine, will be simply connected, with a trivial causal structure. This might be the first step in constructing our mapping functions $\zeta_{\vec{\gamma}}$ that take points from $\mathcal{W}_{\vec{\gamma}}$ onto $\mathcal{W}_{approx}$. (See Fig. 3.)

A. Topology of $\mathcal{Q}^P$

In the simplest case, the boundary $\mathcal{B}_2$ of a 4-geon at a particular instant in time, $\mathcal{O}[\tau]^P$, has the topology of a simply-connected two-sphere, S^2. If we imagine that $\mathcal{Q}^P$ represents the time dependent evolution of the particle from its "creation" to its "destruction," then we might imagine at first that the boundary of $\mathcal{Q}^P$ would have the topology of the sphere cross (a segment of) the real line, $S^2 \times \mathbb{R}$. However, we will postulate that "most of the time," the boundary of $\mathcal{Q}^P$ actually has the topology of a donut, $S^2 \times S$. This is depicted in Fig. 12 A (where the broken lines are the donut.)

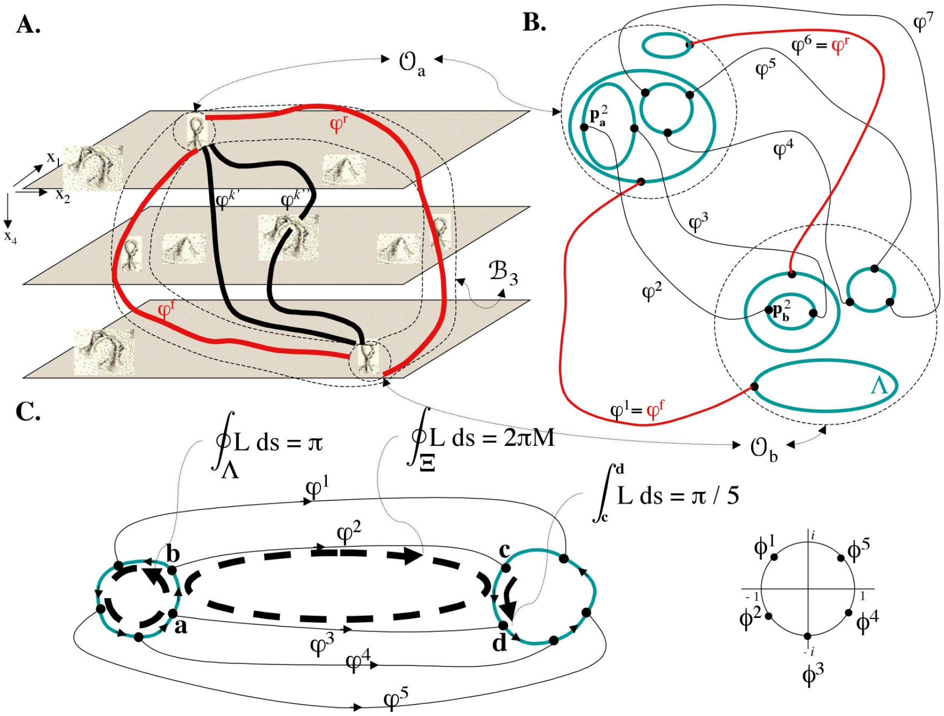

FIG. 12: A. An illustration of a 4-geon. We postulate that 4-geons typically have the topology of a "donut" in 4-space. B. Depiction of the series of loops (green circles) inside a 4-geon, the critical points p on these loops, and the paths that connect these critical points. The number n of critical points on any given loop defines the critical index that characterizes the loop, the points p on the loop, and the paths that connect to it. The two unique forward and reverse paths φ^f and φ^r are postulated to have critical index $n = 1$. C. Left: Paths φ^k are clustered into groups of n paths, where n is the critical index of the path. Depicted is a cluster of 5 paths, i.e., a 5-cluster. Ths integral relations characterizing the paths and loops are discussed in the text. Right: A plot on the complex plane of the amplitudes of the five paths shown on the left. (As in the FPI: $\phi = e^{-iS/\hbar}$ where S is the integral of the lagrangian along the length of the path.) From the integral relations shown on the left, we can derive that any two adjacent paths φ^k and φ^{k+1} in a single n-cluster are $2\pi/n$ out of phase. Thus, for $n > 1$, all of the paths within a given cluster will "cancel each other out" in the FPI method, so the only paths that contribute to the quantum mechanical amplitude are the ones with critical index $n = 1$.

B. Restrictions on the Fundamental Group

Within an arbitrary manifold $\mathcal{W}_{\vec{\gamma}}$, consider any two arbitrary 3-objects $\mathcal{O}_a$ and $\mathcal{O}_b$. Although we will present our discussion below with the intention that they apply to *any* 3-objects $\mathcal{O}_a$ and $\mathcal{O}_b$ *in general*, we will be most concerned with situations in which $\mathcal{O}_a$ and $\mathcal{O}_b$ represent two different time slices of the same 4-geon, e.g.: $\mathcal{O}_a = \mathcal{O}[\tau_1]^P$, $\mathcal{O}_b = \mathcal{O}[\tau_2]^P$.

Within any 3-object $\mathcal{O}_a$, we will define a set P_a of "critical points" p_a^k. We will not present a completed definition of these points – that is work for the future – but will merely assume that one exists, such that these points are able to fulfill the properties discussed below. Consider the set of K paths φ_a^k (defined as in a previous section, that is, as class representatives) connecting $\mathcal{O}_a$ to any other position in spacetime. (Also see the overview of the FPI in the Appendix.) Within $\mathcal{O}_a$, we will assume that each path φ_a^k can be defined to have its endpoint on one of the critical points p_a^k, and that there is a one-to-one correspondence between φ_a^k and p_a^k. Thus, p_a^k is the endpoint of path φ_a^k. (Likewise for b.) In this way, the set of all paths from $\mathcal{O}_a$ to $\mathcal{O}_b$, $\varphi_{a,b}^k$, places a subset of the points p_a^k in $\mathcal{O}_a$ in a one-to-one correspondence with a subset of the points p_b^k in $\mathcal{O}_b$. Given a path $\varphi_{a,b}^k$, which connects the critical point p_a^k in $\mathcal{O}_a$ to p_b^k in $\mathcal{O}_b$, we say that p_a^k is "in correspondence with" p_b^k, or we write: $p_a^k \leftrightarrow p_b^k$. (See Fig. 12 B.)

Within any 3-object $\mathcal{O}_a$, consider the set of loops Λ (maps from the circle into the manifold $\mathcal{O}_a$) that are elements of the fundamental group of $\mathcal{O}_a$. (See the Appendix for a definition of the fundamental group.) As was the case for the critical points above, we will not present a completed definition of these loops – that is work for the future – but will merely assume that one exists, such that these loops may fulfill the properties listed below. We will assume that each critical point p exists on one and only one loop Λ. However, each loop Λ may contain any number of critical points, that is, an integer value number of points in $[1, \infty]$. The number n of critical points on any given loop we will refer to as the "critical index," or simply the "index," of the loop. Furthermore, if a point p exists on a loop with index equal to n, we also assign the value n to be the "critical index," or simply "index," of the point p. Within any object $\mathcal{O}_a$, define χ_a^n as the total number of points p_a^k with index value n.

Given any two points (one in $\mathcal{O}_a$, one in $\mathcal{O}_b$) that are in mutual correspondence, $p_a^k \leftrightarrow p_b^k$, we assume that the index of p_a^k equals the index of p_b^k. Furthermore, we assume that all n of the points p on ony given loop in $\mathcal{O}_a$ will be in correspondence with n points that *all* exist *on a single loop* in $\mathcal{O}_b$. (See Fig. 12 B and C.)

We assume the loops Λ to be oriented, as depicted in Fig. 12 C (the integral on the left). We postulate that the length of each of these loops, as defined by the integral of the lagrangian along the loop, equals π:

$$\oint_\Lambda L_{ab} ds = \pi \tag{1}$$

Furthermore, we postulate that the critical points on this loop are distributed "equidistant" from one another, so that any two *adjacent* critical points will be separated by a distance π/n from one another. (See Fig. 12 C, the integral on the right).

Consider now any closed loop Ξ that is formed by two of these n paths along with segments of the two Λ loops. (See Fig. 12 C, the integral in the middle.) Postulate that the line integral of the lagrangian along this loop Ξ is an integer multiple of 2π:

$$\oint_\Xi L_{ab} ds = 2\pi M \tag{2}$$

where M is some integer.

C. Forward and Reverse paths: φ^f and φ^r

Of all paths $\varphi_{a,b}^k$ from $\mathcal{O}_a$ to $\mathcal{O}_b$, where each of these 3-objects represents two "time slices" through the same particle $\mathcal{Q}^P$, two of these paths are unique in the sense that they can be deformed to stay on the surface $\mathcal{B}_3$ of the 4-geon $\mathcal{Q}^P$ throughout the entire length of the path. Using our postulated model of a 4-geon as having the topology $S^2 \times S$ – that is, the topology of a donut in 4-space – we can think of one of these paths as going "one way around" the 4-geon, and the other path going the "other way around." We label these paths φ^f and φ^r (for forward and reverse paths.) This is depicted in Fig. 12 A.

Recall also that each path $\varphi_{a,b}^k$ is characterized by an index value n that equals the critical index of each of the critical points at the two ends of the path. We will postulate the restriction that the forward and reverse paths, φ^f and φ^r, must each be characterized by critical index $n = 1$.

D. Observation of the state of a particle by an external observer

Recall our earlier discussion (Sec. II M) of the notion that any given internal physical attribute of a particle may or may not be "leaked" into the environment, and that the temporal order with which this information is leaked will have a dramatic effect on the probability of collapse onto some particular state

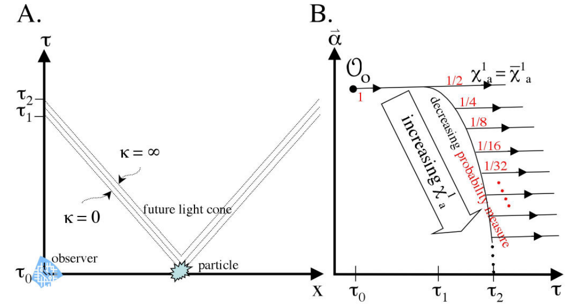

FIG. 13: Observation of the position of a particle. A. A particle leaks information regarding its internal state via emission of a packet of gravitational waves that propagate at speed c. Each wave conveys information regarding the descriptor χ_a^1 of the particle, i.e., that χ_a^1 is greater than or less than the value κ. B. The corresponding tree diagram will result in a high likelihood of collapse onto the lowest possible value of χ_a^1.

of the particle. In this section, we will postulate an order with which information regarding the state χ_a^1 is leaked by a particle into its environment. In the next section, we will see how this postulate plays an important role in our demonstration of the equivalence between the RHF and the FPI.

Recall that χ_a^1 corresponds to the total number of critical points of index one within the three-space $\mathcal{O}_a$, where we typically assume that $\mathcal{O}_a$ matches our definition of a 4-geon: $\mathcal{O}_a = \mathcal{O}_a^P$. Any individual $\mathcal{O}_a^P$ is characterized by the variable χ_a^1. But from the perspective of the observer $\mathcal{O}_o$, the particle may exist in a superposition of different values of χ_a^1, since different $\mathcal{W}_{\vec{\gamma}}$'s in $E_{\vec{\alpha}}^{\mathcal{W}}$ will typically have different representations of this particle. Of all possible values of χ_a^1, denote the *minimum* value of χ_a^1 with a line over the top: $\bar{\chi}_a^1$. Next, define a variable κ. We will assume that the particle "leaks" information in the form of some sort of gravitational waves that contain one bit of information of the following sort: that the value of χ_a^1 is either greater than or less than κ. We furthermore assume that the particle leaks this information as a *series* of gravitational waves – perhaps in a tight packet – and that the value of κ is different for each wave, with the first waves characterized by small κ, and with the value of κ increasing progressively as further waves are emitted.

Let us see what effect this postulate has on the shape of the tree diagram of the observer. Given the discussion of Sec. II M, and considering the restrictions of the chronology protection conjecture (Fig. 13 A), we know that the tree diagram of the observer must take the form depicted in Fig. 13 B. We see quickly that this will have the effect of making collapse onto $\bar{\chi}_a^1$ much more likely than collapse onto higher values.

χ_a^1, by definition, is the sum of $\chi_{a,b}^1$ over all possible values for $b(x,t)$. Therefore, collapse onto $\bar{\chi}_a^1$ implies collapse onto the minimum values of each individual $\chi_{a,b}^1$ (that is, for any given $b(x,t)$). We will likewise denote these variables with a line over the top: $\bar{\chi}_{a,b}^1$.

We will call this postulate the "kernalization" postulate of fundamental particles because of the correspondence (derived

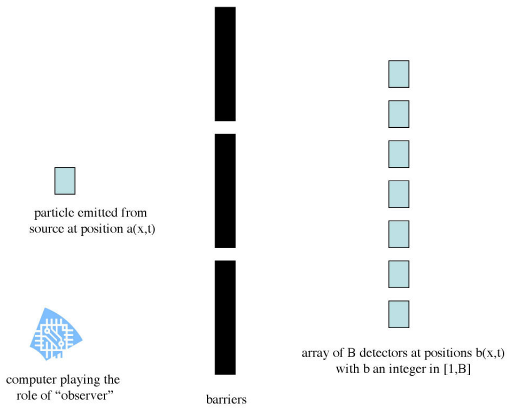

FIG. 14: The "fundamental" quantum mechanical experiment. There is a particle source (on the left), a system of barriers (in the middle), and a series of B detectors (on the right). The computer/observer first records that the experiment is set up as described, and then records the output of the detectors.

below) between $\bar{\chi}^1_{a,b}$ and the absolute value of the total amplitude $\Psi_{a,b}$ for a particle at position $a(x,t)$ to be observed at position $b(x,t)$, which, in the standard terminology of the FPI, is called the "kernel." That is, we will demonstrate below that $\bar{\chi}^1_{a,b} \sim |\Psi_{a,b}|$. We present the kernalization postulate simply because it works, in our overall scheme, of demonstrating compatibility with quantum statistics. This is our sole justification.

IV. MANIFESTATION OF QUANTUM STATISTICS IN THE RHF

It is possible to distill all of quantum mechanics down to a solution to the problem: given a particle that is observed at location $a(x,t)$, calculate the probability that it will be observed at $b(x,t)$. Indeed, the core formalism of the Feynman path integral (FPI) is no more and no less than a solution to this problem. From this starting point, all of quantum mechanics, including the Schrödinger equation, can be derived [1]. We will therefore refer to this problem – which is illustrated in Fig. 14 – as the "fundamental" QM experiment. In this section, we will demonstrate that the RHF and the FPI make the same predictions with respected to the fundamental experiment. Our goal is to demonstrate in explicit terms that the RHF is in fact an alternative formulation of quantum mechanics.

A. The "fundamental" problem

The fundamental experiment is depicted in Fig. 14. It is assumed that a particle is observed in a region of spacetime $a(x,t)$. (This may be accomplished by placing a particle emitter at $a(x,t)$ and establishing a method of verifying successful emission of the particle.) There is an array of N particle detectors labeled $b_i(x,t), i = 1 \ldots N$. At every point in spacetime, we are provided with a potential $V(x,t)$. The black boxes in the figure represent a system of barriers that can be modelled as regions of infinite potential. Although we have depicted the barriers in the configuration of the classical double-slit experiment, they may be distributed in any manner desired. (Likewise, although for the sake of simplicity we depict the detectors to be distributed along a single axis, we could in principle distribute them in any manner desired.) We make the assumption that if a single particle is emitted from the source a, then it will be absorbed by one and only one detector b_i, where b_0 indicates absorption by one of the barriers. The problem is to calculate the probability that a single particle emitted from a will be detected at $b_i, i = 1 \ldots N$ (assuming that it is not absorbed by one of the barriers). We can model a continuous probability distribution by assuming an arbitrarily large N of arbitrarily small detectors b_i. A very brief summary of the FPI technique in the analysis of this experiment may be found in [17] and is also presented in the appendix to this paper. For an outstanding introduction that is more detailed, see Volume III of the Feynman Lectures on Physics, and for a deeper understanding, see his wonderful text [4].

B. Analysis of the fundamental experiment via the FPI

The appendix contains a brief overview of the analysis of the fundamental experiment by the FPI. Let us simply state the equation that tells us, from the framework of the FPI, the relative probability that the particle will be detected at location $b(x,t)$:

$$\mathrm{P_{a,b}} = |\Psi_{a,b}|^2 \tag{3}$$

C. Analysis of the fundamental experiment via the RHF

The RHF analysis of the fundamental experiment must include, of course, an explicit designation of the "observer." In Fig. 14, the observer is depicted as the computer chip at the bottom left of the figure. The physical state of this chip is designated $\mathcal{O}_o = \mathcal{O}_{\vec{\alpha}}$, and we define $E^{\mathcal{W}}_o = E^{\mathcal{W}}_{\vec{\alpha}}$ as the world ensemble relative to $\mathcal{O}_o$ as in Fig. 8 A. We also define a statement X_o = "the experiment is set up as described in Fig. 14 and a single particle has been observed leaving the source at $a(x,t)$," and we assume that X_o is true relative to $\mathcal{O}_o$. Once $\mathcal{O}_o$ is defined, then it is possible in principle to draw the tree diagram and to evaluate *any given statement* X or array of statements X_b at any point along the tree diagram. Thus, we define X_b as above, i. e. X_{b_i} = "the experiment is set up as described in Fig. 14, a single particle has been emitted from the source at $a(x,t)$, and a single particle has been detected at b_i." Using the tree diagram, we may calculate the relative probability that the particle will "collapse" onto any given b-value in the manner illustrated in Fig. 10.

At this point, we have managed to describe, in broad brushstrokes, the fundamental experiment using the language of the

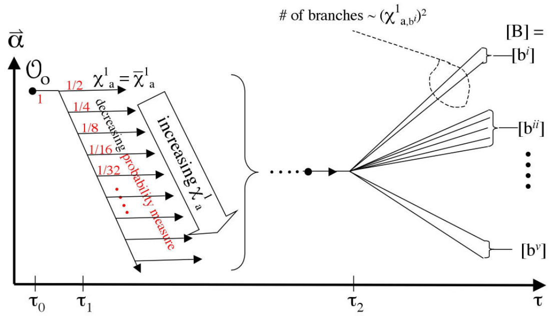

FIG. 15: Three-step structure of the tree diagram. For the fundamental experiment, the tree diagram is broken into three phases. A. First phase: Collapse onto χ^1_a at about τ_1. B. Second phase: Branching, at τ_2. C. Third phase: fusion of branches so that they are grouped by final location b.

RHF – but we are not yet able to come up explicitly with probabilities. To do this, we need to be able to draw the tree diagram. In the next section, we will make use of some of the proposed internal features of particles and will furthermore propose a some features of the tree diagram representation of the fundamental experiment. Our goal will be to demonstrate that the RHF and the FPI technique come up with the same predictions in the analysis of the fundamental experiment. Thus, we will conclude, the RHF makes the same predictions *in general* as quantum mechanics.

D. The tree diagram representation of the fundamental experiment

Using the postulated characteristics of 4-geons in the preceeding section, we will make use of the observable χ^1_a as well as the observable b_i (which corresponds to the particle detector that detects the particle) and use them in the construction of the tree diagram for the fundamental experiment. We postulate that the tree diagram will in general take the form that is depicted in Fig. 15. Let us remind the reader at the outset that we have taken care to postulate a form for the tree diagram that remains consistent with the chronology protection conjecture (CPC). Essentially, this means that the structure of the tree diagram at any given point cannot be a function of any observable that is not accessible to the observer at that point. In particular, we have postulated a model of 4-geons in such a way, and defined χ^1_a in such a way, that we may assume the following: if the presence of the particle at the emitter (that is, at $a(x,t)$) is accessible to the observer, then we may assume that the value of χ^1_a is also accessible to the observer without violation of the CPC. Let us walk through the tree diagram of Fig. 15 step by step.

1. *Step 1*

The first step in the tree diagram corresponds to the observation by the observer that the particle occupies the position $a(x,t)$. (This corresponds to observation of emission of the particle.) Therefore, the first part of the fundamental experiment tree diagram (Fig. 15) is adopted from our earlier discussion of the observation of a particle – see Fig. 13. In Fig. 15, this occurs at (about) time τ_1. Although it is possible to collapse onto *any* value for χ^1_a and $\chi^1_{a,b}$, we know from the kernalization postulate that the most *likely* values are the minimum values, $\bar{\chi}^1_a$ and $\bar{\chi}^1_{a,b}$. Therefore, for the next stages in the tree diagram, we will assume that $\chi^1_a = \bar{\chi}^1_a$ and $\chi^1_{a,b} = \bar{\chi}^1_{a,b}$.

2. *Step 2*

The next phase of the tree diagram begins at the instant τ_2 that the particle is observed by the observer to be in one of the positions $b(x,t)$. At the point in time τ_2, the ensemble $E^{\mathcal{W}}[\tau_2]$ is the union of disjoint subsets $E^{\mathcal{W}}_{f,r}$, where each subset differs from the other subsets in the identity of which two out of the $\bar{\chi}^1_a$ paths constitute the unique forward and reverse paths φ^f and φ^r. We make the postulate that at this point τ_2, the observer collapses onto one of these subsets $E^{\mathcal{W}}_{f,r}$; that is, the observer "observes" which two of the paths are the unique forward and reverse paths. Therefore, there is a unique branch corresponding to each possible pairing. Recall our notation: we have $\bar{\chi}^1_{a,b_i}$ critical index one paths from $a(x,t)$ to $b_i(x,t)$, for a total of $\bar{\chi}^1_a$ critical index one paths from $a(x,t)$ to any detector. Out of these $\bar{\chi}^1_a$ paths, two of them are identified as the unique φ^f and φ^r paths. Of course, φ^f and φ^r must go to the same endpoint $b_i(x,t)$. If there are $\bar{\chi}^1_{a,b}$ paths from a to b, then there will be exactly $\frac{1}{2}\bar{\chi}^1_{a,b} \times (\bar{\chi}^1_{a,b} - 1)$ ways to place these paths into pairs. Therefore, the number of branches to any given b^i in the tree diagram (Fig. 15) will be equal to $\frac{1}{2}\bar{\chi}^1_{a,b} \times (\bar{\chi}^1_{a,b} - 1)$, which for large $\bar{\chi}^1_{a,b}$ is approximately proportional to $\left(\bar{\chi}^1_{a,b}\right)^2$.

3. *Step 3*

In the third and final phase of the tree diagram (Fig. 15), there is a fusion of all branches of the tree diagram leading to the same b.

4. *RHF solution to the fundamental experiment*

According to the postulated form of the tree diagram, the probability that the particle that is emitted from location a will be detected at position b is approximately proportional to $\left(\bar{\chi}^1_{a,b}\right)^2$. We can restate our conclusion as follows: The probability P_{ab} that a particle emitted from position $a(x,t)$ will be detected at position $b(x,t)$ is proportional to the square of $\bar{\chi}^1_{a,b}$, that is:

$$\mathrm{P_{a,b}} \sim \left(\bar{\chi}^1_{a,b}\right)^2 \tag{4}$$

In this manner, we have described in general terms the RHF approach to the fundamental quantum mechanical experiment. As we will see below, the formalism of the FPI makes predictions that are in agreement with the above predictions of the RHF.

E. Correspondence between the RHF and the FPI

We will now attempt to show that the RHF and the FPI will in general make the same predictions with respect to the fundamental experiment. Since the RHF predicts that $\mathrm{P_{a,b}} \sim \left(\bar{\chi}^1_{a,b}\right)^2$, and the FPI predicts that $\mathrm{P_{a,b}} \sim |\Psi_{a,b}|^2$, then our goal is to demonstrate that $\bar{\chi}^1_{a,b} \sim |\Psi_{a,b}|$, which we will do shortly.

Let us recall that the FPI and the RHF techniques each begin by enumerating the set of "all possible paths" from source a to detectors b, and that in each case, we typically approximate the "K most representative" paths. As discussed previously, we imagine that the FPI and the RHF make use of the *same* paths φ.

Recall that according to the RHF technique, each path from a to b is associated with one loop in a as well as one loop in b. Each of these loops is characterized by the same critical index n, which equals the number of critical points on the loop, and thus equals the number of paths associated with that loop. Using the relationships postulated in Sec. III and depicted in Fig. 12, we derive the following theorem:

Theorem 1. *Any two adjacent paths φ^k and φ^{k+1} that are coassociated with the same loop (whose critical index is n) will be $2\pi/n$ out of phase.*

Proof. By "out of phase" we mean that the amplitudes of the two paths are separated by an angle $2\pi/n$ when plotted on a phase diagram. See Fig 12 C. We are integrating the lagrangian around the thick green loop in the middle of the diagram, breaking it into four segments at the points a, b, c, and d. For simplicity, let us set $M = 1$. S^k indicates the action of the k-th path. (See the appendix section on the FPI for the definition of the action.)

$$\oint_{\Xi} L_{ab} ds = 2\pi$$

$$\int_a^b L_{ab} ds + \int_b^c L_{ab} ds + \int_c^d L_{ab} ds + \int_d^a L_{ab} ds = 2\pi$$

$$\pi/n + S^k + \pi/n - S^{k+1} = 2\pi$$

$$S^k - S^{k+1} = 2\pi - \frac{2\pi}{n}$$

$$S^k - S^{k+1} = 2\pi \left(1 - \frac{1}{n}\right)$$

□

Using this result, let us analyze the FPI formula for the relative probability of detection of the particle at position b.

$$\Psi_{a,b} = \sum \phi_{a,b} \tag{5}$$

The sum on the right can be divided into the paths with critical index n equal to one, and those paths with critical index n greater than one:

$$\Psi_{a,b} = \sum_{(n=1)} \phi_{a,b} + \sum_{(n>1)} \phi_{a,b} \tag{6}$$

Given the theorem above, we see that the term on the right equals zero:

$$\sum_{(n>1)} \phi_{a,b} = 0 \tag{7}$$

so we are left with:

$$\Psi_{a,b} = \sum_{(n=1)} \phi_{a,b} \tag{8}$$

In other words, the Feynman amplitude $\Psi_{a,b}$ can be calculated using the RHF formalism by neglecting all paths with critical index greater than one, and by simply summing the amplitudes of only those paths that are associated with a loop with critical index $n = 1$.

Recall that the first phase in the tree diagram of the fundamental quantum mechanical experiment (Fig. 15) has the effect of minimizing $\chi^1_{a,b}$. Looking three equations back, that means that we have collapsed onto those manifolds that have stuffed as many paths as possible into the right hand summation term (for $(n > 1)$), leaving as few as possible into the left hand summation term (for $(n = 1)$). Assuming that this is the case, we must have that each of the paths with $n = 1$ has the same phase, in which case the *number* of individually clustered paths (i.e., $\bar{\chi}^1_{a,b}$) will be proportional to the absolute value of the sum of the amplitudes of each of the $n = 1$ paths: $\bar{\chi}^1_{a,b} \sim |\sum_{(n=1)} \phi_{a,b}|$. Thus we can conclude that $\bar{\chi}^1_{a,b} \sim |\Psi_{a,b}|$, which – as we stated at the beginning of this section – indicates that the FPI and the RHF will in general make identical predictions in the analysis of the fundamental quantum mechanical experiment. This is what we have set out to prove.

F. Analysis of a sample problem

We will now consider the experimental setup of Fig. 16 A and perform an analysis using the formalism of the RHF and the formalism of the FPI. Our goals will be to first to all to clarify the concepts introduced above, and second, to recapitulate the notion that the FPI and the RHF make the same predictions. The ultimate conclusion, therefore, is that the basic framework of the RHF can in fact be constructed in such a fashion that it gives rise to quantum statistics.

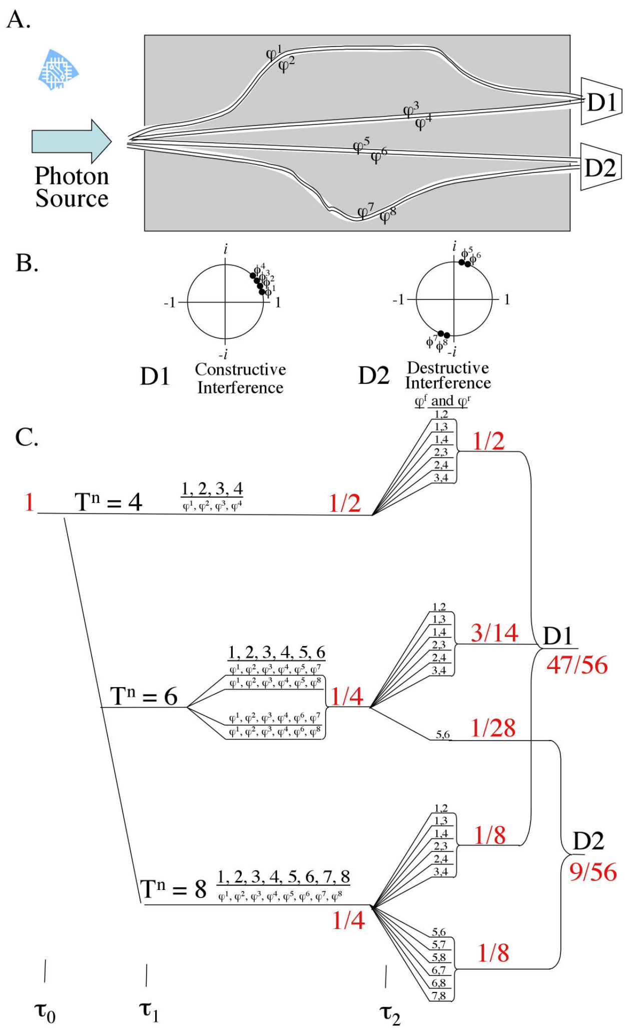

FIG. 16: Sample Experiment. A. Experimental setup. B. Analysis by the FPI. There is nearly 100 percent chance of detection at D1. C. Tree diagram of the RHF. Depicted in red are the relative probabilities associated with various points on the tree diagram, which can be calculated from the tree diagram itself via application of the random walk hypothesis. By definition, the probability measure at the origin of the tree diagram is normalized to unity. For our choice to approximate 2 representative paths through each fiberoptic cable, we predict an approximately 84 percent and 16 percent chance of detection at D1 or D2, respectively. As we allow the number of representative paths to increase, the predicted probabilities will approach 100 percent and 0 percent, respectively. Thus, as the number of approximated paths increases, the RHF prediction becomes ever nearer to the FPI prediction.

1. *Experimental Setup*

Consider the experimental setup of Fig. 16 A. On the left, there is a photon emitter that emits single photons. On the right there are two photon detectors, $D1$ and $D2$. Between source and detectors there are several fiberoptic cables, two leading to $D1$ and two leading to $D2$, which are embedded in an opaque material that serves as a photon barrier.

Using the technique of the FPI, let us consider, say, the 2 most representative paths through each cable, for a total of 8 paths from photon source $a(x,t)$ to photon detectors $b_1(x,t)$ and $b_2(x,t)$. We will label these paths φ^k with $k = 1, 2, \ldots, 8$ as indicated in the figure. Let us assume that the action S^k of each of the 2 paths through any individual cable are approximately equal. Therefore, a plot of the 2 amplitudes ϕ^k of any given cable on a phase diagram will each fall at about the same place on the unit circle, as depicted in Fig. 16 B. Furthermore, we will assume that the lengths of the cables are such that the two cables leading to $D1$ are in phase with one another, while the two cables leading to $D2$ are 180 degrees out of phase with one another. These relationships are depicted in Fig. 16 B.

2. *Analysis by the FPI*

According to the FPI technique, the *total* amplitude for any given photon to arrive at $D1$ is the sum of the 4 amplitudes leading to $D1$; likewise for $D2$. Since the four paths leading to detector $D1$ (located at spacetime location b_1) are all in phase with one another, while the paths leading to $D2$ are out of phase with one another, we can easily see that $|\Psi_{a,b_1}|$ will be much larger that $|\Psi_{a,b_2}|$. Thus, the probability of detection at $D1$, $|\Psi_{a,b_1}|^2$, will be close to 100 percent, while the probability of detection at $D2$, $|\Psi_{a,b_2}|^2$, will be close to zero percent. We say using the standard terminology that the two fiberoptic cables leading to $D1$ interfere *constructively* with one another, whereas the two cables leading to $D2$ interfere *destructively*.

3. *Analysis by the RHF*

Let us now analyze this setup using the formalism of the RHF. The first step is to identify the observer $\mathcal{O}_o = \mathcal{O}_{\vec{\alpha}}$, which we have drawn as the computer recording device to the left of the experimental apparatus (Fig. 16 A). We define $\mathcal{Q}^P$ as the 4-object that represents our particle (the photon). We also define $\mathcal{O}^P_a$ as the 3-object that describes the state of the particle as it is emitted from the emitter at spacetime location $a(x,t)$. Likewise $\mathcal{O}^P_b$ is the 3-object that describes the state of the particle at spacetime location $b(x,t)$, which we also identify with the detector $D1$ or $D2$. As was the case with the FPI analysis, we consider the 2 most representative paths through each fiberoptic cable, for a total of 8 most representative paths. Each element $\mathcal{W}_{\vec{\gamma}}$ of the ensemble $E^{\mathcal{W}}_o$ therefore has an internal representation of *all eight paths*; these are in one-to-one correspondence with the eight paths in $\mathcal{W}_{approx}$. So we can think of these as being the "same" eight paths in each and every $\mathcal{W}_{\vec{\gamma}}$ in $E^{\mathcal{W}}_o$. Each individual $\mathcal{W}_{\vec{\gamma}}$ also has an internal representation of the photon $\mathcal{Q}^P$. The details of this representation, however, differ from $\mathcal{W}_{\vec{\gamma}}$ to $\mathcal{W}_{\vec{\gamma}}$. In particular: For any given individual $\mathcal{W}_{\vec{\gamma}}$, there is a particular value for χ^1_a (as well as χ^1_{a,b_1} and χ^1_{a,b_2}), a particular identification of the final destination of the particle (b_1 or b_2), and a particular identification of which two of the eight paths correspond to the forward and reverse paths φ^f and φ^r.

Next, we draw the tree diagram that corresponds to the experiment. This is depicted in Fig. 16 C. At τ_0, the observable χ^1_a (which is the sum of $\chi^1_{a,b_1} + \chi^1_{a,b_2}$) is in a superposition of all possible values (which we discuss below). Likewise, the observable b is in a superposition of the two possible final values: b_1 and b_2.

Let us consider the different possible values of χ^1_{a,b_1}. Since all 4 paths leading to $D1$ are in phase with one another, there is no way for more than one of their critical points p_a at the particle source $a(x,t)$ to be coassociated on a single loop Λ. (Recall that the n paths on a loop Λ must be $2\pi/n$ out of phase with one another.) Thus, each of these 4 paths must be separated into four separate loops, each of critical index equal to 1. Thus, $\bar{\chi}^1_a$ (the minimum value of χ^1_a) must be at least four, corresponding to the situation that there are four loops with critical index 1 each of which contains one critical point p_a that corresponds to one of the four paths leading to $D1$.

It is likewise possible for the 4 paths leading to $D2$ to be each individually associated with their own 4 separate loops, each with critical index equal to 1. For any manifold $\mathcal{M}_{\vec{\gamma}}$ for which this is the case, we will have $\chi^1_a = 8$. In fact, this is our maximum value for χ^1_a.

However, since the 4 paths leading to $D2$ are out of phase with one another, it is possible that some of these paths will be pairwise associated with loops at $a(x,t)$ whose critical index equals two. For example, φ^6 and φ^8, which are 180 degrees out of phase with one another (see diagram), may be coassociated with a single loop in $a(x,t)$ with critical index $n = 2$ – and also with a single loop in $b_2(x,t)$ with critical index $n = 2$ – leaving the other six paths $\varphi^1, \varphi^2, \varphi^3, \varphi^4, \varphi^5, \varphi^7$ to be individually associated with six separate loops, each of critical index equal to one. In this case, we have $\chi^1_a = 6$ (also, incidentally, $\chi^2_a = 2$). (The diagram also depicts three other combinations of paths such that $\chi^1_a = 6$.)

The first phase of the tree diagram, therefore, corresponds to "collapse" onto one of the three possible values of χ^1_a. According to the way we have postulated that the tree diagram is constructed, the lower values of χ^1_a are more likely than the higher values. In this particular case, there is a 50 percent chance of collapse onto its minimum value, $\chi^1_a = \bar{\chi}^1_a = 4$; a 25 percent chance of collapse onto $\chi^1_a = 6$; and 25 percent chance of collapse onto $\chi^1_a = 8$. This can be deduced from the tree diagram as depicted in Fig. 16 C, keeping in mind the random walk hypothesis.

Consider now the case of "collapse" onto the branch corresponding to $\chi^1_a = 6$. The ensemble corresponding to this point on the tree diagram, $E^{\mathcal{W}}[\tau_1]$, consists of elements $\mathcal{W}_{\vec{\gamma}}$ in which there are a total of 6 paths whose critical index equals one. We may label these paths 1 through 6, as depicted in the diagram. For any given $\mathcal{W}_{\vec{\gamma}}$ in $E^{\mathcal{W}}[\tau_1]$, two of these 6 paths correspond to the two unique forward and reverse paths, φ^f and φ^r. However, the identity of these two unique paths will typically differ from one $\mathcal{W}_{\vec{\gamma}}$ to the next.

How many different combinations of ways are there to pick two of these 6 paths as the unique forward and reverse paths? In any given $\mathcal{W}_{\vec{\gamma}}$, φ^f and φ^r must end at the same endpoint, that is, b_1 or b_2. (In other words, we cannot have, for example, φ^f going to b_1 and φ^r going to b_2.) Since there are 4 paths leading to $D1$, there are $4 * 3/2 = 6$, or approximately $\frac{1}{2}(\chi^1_{a,b_1})^2$, combinations of ways for φ^f and φ^r to end at $D1$. But since there are only two paths leading to $D2$, there is only one way for φ^f and φ^r to end at $D2$. These different numbers of combinations are reflected in the numbers of branches of the tree diagram at τ_2.

For each of the other branches corresponding to different values of χ^1_a, the branch diagram at τ_2 is constructed in a similar manner as described above. Through repeated application of the random walk hypothesis, we see that the overall probability of collapse onto b_1 is much greater than the overall probability of collapse onto b_2: 84 percent versus 16 percent. Of course, the notion that there are only 2 paths through each of the fiberoptic cables is an approximation technique; in fact, there are an infinite number of paths through each cable. We can easily see that as we approximate more and more paths through each cable, the overall probability of collapse onto b_1 approaches 100 percent, while the overall probabilty of collapse onto b_2 approaches 0 percent, in agreement with the Feynman path integral method (Sec. IV F 2).

In conclusion to this section, we have shown by example how the RHF handles constructive as well as destructive interference, and we have shown that the RHF makes the same predictions (at least in this particular example) as the FPI technique. The basic methodology outlined in general terms above, and illustrated in these example problems, may be implemented in the analysis of the fundamental quantum mechanical experiment. As we argued above, the RHF and the FPI will in the approximation make the same predictions. And since the fundamental experiment can be viewed to form the basis of any experiment whatsoever, we conclude that the RHF and the FPI will make the same predictions with respect to any experiment whatsoever. Therefore, we conclude that the RHF qualifies as an alternative formulation of quantum mechanics, one that is founded on a different set of starting assumptions than any of the other contemporary formulations of quantum mechanics. We hope to make it clear in the discussion, where we summarize these basic starting assumptions, that the RHF does not simply assume quantum statistics outright. Rather, it assumes classical statistics, and – rather amazingly – manages at the same time to make predictions that are in agreement with quantum mechanics.

V. DISCUSSION

The relative histories formulation (RHF) is proposed as an alternative mathematical foundation for quantum mechanics. The primary purpose of this introductory paper is to present the basic mathematical elements of the RHF and to demonstrate that the RHF constitutes an alternative formulation of quantum mechanics. Our method was to show that the RHF is equivalent to the Feynman path integral – i.e., that they make, *in general*, the same predictions with respect to the "fundamental" experiment. Since the FPI provides a basis for the derivation of the Schrödinger equation, we conclude that the RHF makes the same predictions as quantum mechanics in general.

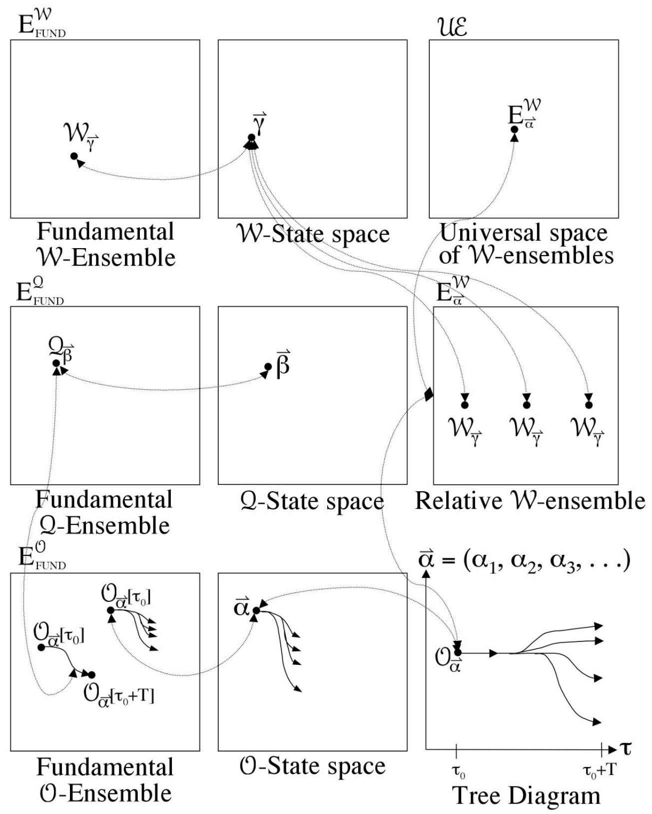

FIG. 17: Overview of the various spaces that are used in the RHF. The fundamental $\mathcal{W}$-ensemble contains one "copy" of every possible world manifold. A "relative $\mathcal{W}$-ensemble" is defined in terms of the object manifold $\mathcal{O}_{\vec{\alpha}}$. In this case, any given element $\mathcal{W}$ will typically be represented more than once. $\mathcal{UE}$ is a Hilbert space. See [20] for a further discussion of the properties of these spaces.

The primary motivation for the development of the RHF is that it lends itself to a constellation of interpretational features that are not found in any other formulation of quantum mechanics. These interpretational issues are the subject of a companion paper [19]. Perhaps the most remarkable feature of the RHF is the fact that it gives rise to quantum statistics without simply adopting quantum statistics outright as one of its central assumptions. Rather, our definition of a probability measure is fully consistent with pre-quantum notions of how probability works. According to our scheme, the time dependent evolution of any arbitrarily chosen three dimensional object ($\mathcal{O}_o = \mathcal{O}_{\vec{\alpha}}$) is modelled as a random walk through state space ($\mathcal{O}$-State space) in much the same way that a dust particle would meander randomly through the air. To understand how the RHF is also consistent with quantum mechanics – which, of course, is *not* consistent with pre-quantum notions of probability – we make the distinction between "how probability works" in the description of the evolution of the observer through $\mathcal{O}$-State space – which is classical – versus "how probability works" in the description of the evolution of a particle (not the observer) in $\mathcal{W}_{approx}$. This topic is addressed in some detail in [19].

So let us at this point present an overview of the postulates of the RHF, so that we may emphasize that we do not simply *assume* quantum statistics. We will group them into several broad categories: (1) First is the *Central Postulate of the RHF*. The physical state of any object is represented by a 3-object manifold $\mathcal{O}$, and its time dependent evolution is modelled as a flow down one branch of the tree diagram. (2) The probability of flow down any particular branch of the tree diagram is governed by the *random walk hypothesis*, according to which each of N branches of a branch point on a tree diagram tree is equally likely. (3) We assume that Hawking's *chronology protection conjecture* applies on the large scale but not on the small scale. This postulate is more or less equivalent to the postulate that spacetime is warped and twisted on the small scale, but basically flat at the large scale. (4) Related to the CPC is the *path equivalency hypothesis*, according to which the action of a path φ in $\mathcal{W}_{\vec{\gamma}}$ can be closely approximated by the action of its image in $\mathcal{W}_{approx}$ obtained by the mapping $\zeta_{\vec{\gamma}}$. (5) Particles are modelled as 4-geons and have the topological characteristics postulated in this paper. These characteristics include (6) the notion that a 4-geon $\mathcal{Q}^P$ "usually" has the topology of a donut in four-space, that is, $S^2 \times S$ (as depicted in Fig. 12). We also make use of (7) the restrictions on the fundamental group, according to which the line integral along certain loops are integral multiples of π, as outlined in Sec. III B. Finally, we make use of (8) the *kernalization postulate*, which postulates a restriction on the order with which information regarding the internal characteristics of a particle are "leaked" into the environment.

We have attempted to provide some intuitive justification for these postulates, where possible, as they were introduced throughout this paper. In a companion paper [19], some of the motivations behind these postulates will be discussed in the context of the established formulations and interpretations of quantum mechanics. Certainly, work remains to be done to translate the above postulates into a form that is more mathematically rigorous than has been presented in this paper. Hopefully it is clear from the exposition of the desire to formulate, or perhaps derive, at least a subset of the above postulates from more fundamental properties of the various spaces and manifolds that form the basis of the RHF. (An overview of these spaces is presented for reference in Fig. 17.) The details of our postulates, notably the latter ones (four through eight), may certainly undergo some degree of modification as this derivation is attempted.

A. Future work

Much work needs to be done regarding the description of the various mathematical entities that are used in the RHF. One restriction that we have placed over our manifolds $\mathcal{W}_{\vec{\gamma}}$ – although we have not made use of this restriction in the current work – is that they must be governed by Einstein's equation. In this way, we may ensure compatibility with General Relativity. What else can we say about our manifolds and their spaces? For example: what is the fundamental group of the space of manifolds? Will it be possible to present a succinct governing equation, GE, from which many of the above postulates may be derived? Might there perhaps be a relationship between this hypothetical GE and Einstein's equation itself? Might GE, in fact, be *equivalent* to Einstein's equation? This would be an exciting discovery, although we are currently unable to provide any argument one way or another.

These questions are the subject of a companion paper [20] (in progress). We anticipate that this further description will most naturally be performed using Morse theory and other mathematical tools that currently form the basis of string theory. Therefore, we anticipate that the RHF may in fact provide a physical foundation for string theories, which heretofore have not been firmly grounded in any set of physical postulates. Unlike string theory, the RHF does not postulate the existence of any extra "unphysical"dimensions beyond the four dimensions of ordinary experience. Most string theories, on the other hand, make use of a ten (or eleven) dimensional state space. In the RHF, the state space of the observer is infinite dimensional (since $\vec{\alpha}$ is an infinite-dimensional vector). Interestingly, as is discussed in [20], each of the components of $\vec{\alpha}$ can be expressed in terms of partial derivatives of the ten-dimensional metric g_{ab}, thus suggesting some sort of connection between the ten dimensions of string theory and the ten dimensions of the metric tensor.

The RHF furthermore suggests the need for further exploration of the 4-geon model of fundamental particles. The present work, for example, does not address the difference between fermions and bosons, nor of the difference between particles with different spin structure. We note that Hadley has used his 4-geon model of particles to propose a model [11] of spin one-half particles in a manner that is consistent with the general scheme of the RHF.

Acknowledgments

We wish to acknowledge Ed Huff, Mark Hadley, zapperz, his evil twin alacrity, Tom Gutierrez, Scott C. Johnson, David Charles Hobby, vznuri, scerir, Travis Norsen, Eric Dennis, and others for their extremely helpful discussions.

APPENDIX A: DEFINITIONS AND THEOREMS

1. Manifold

Essentially, a manifold is a continuous space which looks locally like Euclidean space [16] (page 151). Our definition of a manifold is reproduced in almost exact form from Wald, chapter 2 [22]. First, we remind the reader that an *open ball* in $\mathbb{R}^n$ of radius r centered around point $y = (y^1, \ldots, y^n)$ consists of the points x such that $|x - y| < r$, where

$$|x-y| = \left[\sum_{\mu=1}^{n} (x^\mu - y^\mu)^2\right]^{1/2} \tag{A1}$$

An *open set* in $\mathbb{R}^n$ is any set that can be expressed as a union of open balls.

Basically, a *manifold* is a set made up of pieces that "look like" open subsets of $\mathbb{R}^n$ such that these pieces can be "sewn together" smoothly. More precisely, an *n-dimensional, C^∞, real manifold* is a set together with a collection of subsets $\{O_\alpha\}$ satisfying the following properties:

(1) Each $p \in M$ lies in at least one O_α, i.e., the $\{O_\alpha\}$ cover M.

(2) For each α, there is a one-to-one, onto, map $\psi_\alpha : O_\alpha \to U_\alpha$, where U_α is an open subset of $\mathbb{R}^n$.

(3) If any two sets O_α and O_β overlap, $O_\alpha \cap O_\beta \neq \emptyset$ (where $\emptyset$ denotes the empty set), we can consider the map $\psi_\beta \circ \psi_\alpha^{-1}$ (where $\circ$ denotes composition) which takes points in $\psi_\alpha [O_\alpha \cap O_\beta] \subset U_\alpha \subset \mathbb{R}^n$ to points in $\psi_\beta [O_\alpha \cap O_\beta] \subset U_\beta \subset \mathbb{R}^n$. We require these subsets of $\mathbb{R}^n$ to be open and this map to be C^∞, i.e., infinitely continuously differentiable. (Since we are dealing here with maps of $\mathbb{R}^n$ into $\mathbb{R}^n$, the advanced calculus notion of C^∞ applies.)

2. Diffeomorphism

Consider manifolds M and M' and a map $f : M \to M'$. If f is C^∞, one-to-one, onto, and has C^∞ inverse, then f is called a diffeomorphism and M and M' are said to be diffeomorphic. If no such f exists, then M and M' are not diffeomorphic. Diffeomorphic manifolds have identical manifold structure. (Summarized from [22], pages 13-14.)

3. Simply connected, multiply connected, pathwise connected

(Adapted from reference [23].)

A space D is connected iff any two points can be connected by a curve lying wholly within D.

A topological space X is pathwise connected iff for every two points $x, y \in X$, there is a continuous function from $[0, 1]$ to X such that $f(0) = x$ and $f(1) = y$.

For manifolds (although not for every type of space), being connected and being pathwise-connected are equivalent.

A pathwise-connected domain is said to be simply connected (also called 1-connected) if any simple closed curve can be shrunk to a point continuously in the set. If the domain is connected but not simply, it is said to be multiply connected.

4. Bounded and unbounded manifolds

(Adapted from reference [23].)

A set in a metric space (X, d) is bounded if it has a finite generalized diameter, i.e., there is an $R < \infty$ such that $d(x, y) \leq R$ for all $x, y \in X$. A set in $\mathbb{R}^n$ is bounded if it is contained inside some ball $x_1^2 + \ldots + x_n^2 \leq R^2$ of finite radius R.

5. Time orientable

6. Equivalence class, class representatives

(Adapted from reference [23].)

An equivalence class is defined as a subset of the form $\{x \in X : xRa\}$, where a is an element of X and the notation "xRy" is used to mean that there is an equivalence relation between x and y. It can be shown that any two equivalence classes are wither equal or disjoint, hence the collection of equivalence classes forms a partition of X. For all $a, b \in X$, we have aRb iff a and b belong to the same equivalence class.

A set of *class representatives* is a subset of X that contains exactly one element from each equivalence class.

7. Fundamental Group

(Adapted from reference [23].)

The fundamental group of an arcwise-connected set X is the group formed by the sets of equivalence classes of the set of all *loops*, i.e., paths with initial and final points at a given basepoint p, under the equivalence relation of *homotopy*. Two mathematical objects are said to be *homotopic* if one can be continuously deformed into the other.

APPENDIX B: SUMMARY OF THE FPI

In this section, we present a compact overview of the basic steps in the analysis of a typical quantum mechanical experiment via the formalism of the FPI. These steps are illustrated in Fig. 18. We summarize from Ref. [4].

Suppose a particle starts at point $a(x,t)$. We wish to calculate the probability that our measurement device will find it at point $b(x,t)$.

Step 1: Draw every path φ from a to b. In general, there will be a truly infinite number of paths. However, it is possible to approximate the path integral technique using the "K most representative" paths, where K is a (suitably large but finite) integer. So when we approximate, the paths are labelled with the variable k, where $k = 1, 2, \ldots, K$. In the general case (with an infinite number of paths), we use the continuous variable z instead of the discrete variable k.

Step 2: For each path $\varphi^k_{a,b}$, calculate the lagrangian $L^k_{a,b}$:

$$L^k_{a,b}(t) = KE(x,t) - PE(x,t) \tag{B1}$$

where KE is the kinetic energy and PE is the potential energy of the particle.

Step 3: For each path $\varphi^k_{a,b}$, calculate the classical action $S^k_{a,b}$ as the integral of the lagrangian along the path:

$$S^k_{a,b} = \int L^k_{a,b} dt. \tag{B2}$$

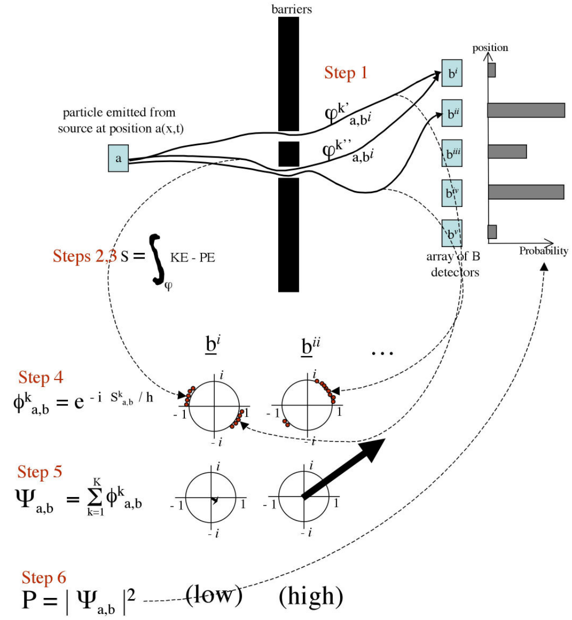

FIG. 18: Analysis of a generic quantum mechanical experiment by the FPI. We wish to calculate the probabilities that a particle that starts at position $a(x,t)$ will be detected at position $b(x,t)$. See the text for details. Several different particle paths from source to detector are illustrated. We have also represented the amplitudes ψ of $K = 10$ "representative" paths by plotting them as closed red dots on the unit circle in the complex plane. The total amplitude Ψ to be detected at a particular spot b on the detector is the sum of each of the individual amplitudes. b^i is an example of destructive interference; b^{ii} is an example of constructive interference.

Step 4: For each path $\varphi^k_{a,b}$, calculate the amplitude $\phi^k_{a,b}$:

$$\phi^k_{a,b} = Ae^{-\imath S^k_{a,b}/\hbar} \tag{B3}$$

where A is a normalization constant.

Step 5: Calculate the wavefunction $\Psi_{a,b}$ that the particle will be detected at b:

$$\Psi_{a,b} = \sum_{k=1}^{K} \phi^k_{a,b} \tag{B4}$$

Step 6: Calculate the relative probability $P_{a,b}$ that the particle will be detected at b:

$$\mathrm{P_{a,b}} = |\Psi_{a,b}|^2 \tag{B5}$$

Step 7: For each individual location $b(x,t)$ on the detector, repeat steps 1 through 6.

Summary:

$$\mathrm{P_{a,b}} = |A|^2 \left| \sum_{k=1}^{K} e^{(-\imath/h)\int_{k^{th}path} KE(x,t)-PE(x,t)dt} \right|^2 \quad \text{(B6)}$$

If we let the sum become an integral (so the paths are labelled by z instead of k), we have

$$\mathrm{P_{a,b}} = |A|^2 \left| \int_{-\infty}^{\infty} e^{(-\imath/h)\int_{z^{th}path} KE(x,t)-PE(x,t)dt} dz \right|^2$$

[1] Egli, Christian. Feynman Path Integrals in Quantum Mechanics. http://www.hedweb.com/manworld.htm http://courses.physics.kth.se/5A1381/reports/egli.pdf

[2] Everett, Hugh. "Relative State" Formulation of Quantum Mechanics. Reviews of Modern Physics. Vol 29, No 3. (July 1957) 454-462.

[3] Feynman, RP, Leighton, RB, and Sands, ML. The Feynman Lectures on Physics. Addison-Wesley Publishing Co., Inc., Reading, MA. 1963.

[4] Feynman, RP and Hibbs, AR. **Quantum Mechanics and Path Integrals**, (Boston: McGraw-Hill, 1965).

[5] L.J. Garay, Int. J. Mod. Phys. A10, 145 (1995)

[6] L.J. Garay. Thermal properties of spacetime foam. gr-qc/9806047; Phys. Rev. D58, 124015 (1998)

[7] Hadley, Mark J. A Gravitational Theory of Quantum Mechanics. (Doctoral thesis) http://www.warwick.ac.uk/ phsem/

[8] Hadley, Mark J. The Logic of General Relativity Derived from Classical General Relativity. Foundations of Physics Letters Vol 10, No.1, (1997) 43-60.

[9] Hadley, Mark J. Topology change and context dependence. International Journal of Theoretical Physics Vol. 38 (1999) 1481-1492

[10] Diemer, T. and Hadley, Mark J. Charge and the Topology of Spacetime. Class. Quantum Grav. 16 No 11 (November 1999) 3567-3577

[11] Hadley, Mark J. Spin half in classical general relativity. Class. Quantum Grav. 17 No 20 (21 October 2000) 4187-4194

[12] Hadley, Mark J. The orientability of spacetime. Class. Quantum Grav. 19 (August 2002) 4565-4571

[13] Spacetime foam. S.W. Hawking, Nucl. Phys. B144, 349 (1978). The Einstein Lagrangian allows large fluctuations of the metric and topology of spacetime on scales of order of the Planck length. This foam-like structure can be described by introducing a term as a Lagrange multiplier for the 4-volume. The dominant contribution to the path integral seems to come from metrics which have about one unit of topology per Planck volume.

[14] S. W. Hawking, Phys. Rev. D 46, 603 (1992); M. J. Cassidy and S. W. Hawking, hep-th/9709066.

[15] Li-Xin Li and J. Richard Gott III. Self-Consistent Vacuum for Misner Space and the Chronology Protection Conjecture. PRL. 80 (14), 6 April 1998 2980-2983.

[16] Schutz, Bernard F. A first course in general relativity, (Cambridge, United Kingdom: Cambridge University Press, 1985).

[17] Styer, Daniel F. et al. Nine formulations of quantum mechanics Am. J. Phys. 70 (3), March 2002. 288-297.

[18] Strayhorn, WD. The Relative Histories Formulation of Quantum Mechanics I: Introduction.

[19] Strayhorn, WD. The Relative Histories Formulation of Quantum Mechanics II: Interpretational Issues.

[20] Strayhorn, WD. The Relative Histories Formulation of Quantum Mechanics III: Further Mathematical Development. (Work in progress)

[21] Thorne, KS. **Black Holes and Time Warps: Einstein's Outrageous Legacy**. W.W. Norton and Company. New York, 1994.

[22] Wald, Robert M. **General Relativity** (Chicago: University of Chicago Press, 1984).

[23] http://mathworld.wolfram.com/AnalyticContinuation.html http://mathworld.wolfram.com/TaylorSeries.html http://mathworld.wolfram.com/Loop.html

[24] "A more detailed scrutiny of a surface might disclose that what we had considered an elementary piece in reality has tiny handles attached to it which change the connectivity character of the piece, and that a microscope of ever greater magnification would reveal ever new topological complications of this type, *ad infinitum*. The Riemann point of view allows, also of real space, topological conditions entirely different from those realized by Euclidean space. I believe that only on the basis of the freer and more general conception of geometry which had been brought out by the development of mathematics during the last century, and with an open mind for the imaginative possibilities which it has revealed, can a philosophically fruitful attack upon the space problem be undertaken." Taken from Weyl, H., 1949, *Philosophy of Mathematics and Natural Science*, Princeton University Press, Princeton, N.J. Quoted also in MTW, Box 8.5.

[25] J.A. Wheeler, Ann. Phys. 2, 604 (1957); J.A. Wheeler, Geometrodynamics (Academic Press, London, 1962); J.A. Wheeler, in Relativity, Groups and Topology, edited by B.S. and C.M. DeWitt (Gordon and Breach, New York, 1964).